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Electrostatic clutch for power tool

Abstract

A power tool includes an electrostatic clutch assembly disposed between a motor and an end effector. A control circuit causes a first voltage to be applied to a first electrode of the electrostatic clutch assembly and a second voltage to be applied to a second electrode of the electrostatic clutch assembly, the first voltage being different from the second voltage, generating a first attractive force between the first and second electrodes, causing a shaft and a clutch housing to be coupled to each other when a torque on one of the shaft or the clutch housing is less than or equal to a first threshold value, and allowing the one of the shaft or the clutch housing to rotationally slip relative to the other of the shaft or the clutch housing when the torque on the one of the shaft or the clutch housing exceeds the first threshold value.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This patent application is a continuation of International Application No. PCT/US2023/073828, entitled “ELECTROSTATIC CLUTCH FOR POWER TOOL”, filed Sep. 11, 2023, which claims priority to U.S. Provisional Patent Application Ser. No. 63/406,063 (“the '063 Provisional Patent Application”) entitled “ELECTROSTATIC CLUTCH FOR POWER TOOL”, filed Sep. 13, 2022. The entireties of these applications are incorporated herein by reference.

FIELD

(1) The present patent application relates to power tools and electrostatic clutches/mechanisms for power tools.

BACKGROUND

(2) Many power tools, such as power drills, power drivers, power fastening tools and/or other power tools, have a mechanical clutch that interrupts power transmission to the output spindle/shaft when the output torque exceeds a threshold value of a maximum torque. U.S. Pat. No. 9,494,200, which is incorporated by reference in the patent application in its entirety, provides an exemplary prior art mechanical clutch. Such a mechanical clutch is a purely mechanical device that breaks a mechanical connection in the transmission to prevent torque from being transmitted from the motor to the output spindle/shaft of the power tool. Clutches or slip clutches are generally used in the power tools to provide torque limited application at the working bit. Traditional slip clutches have been executed mechanically with balls, springs, and clutch plates. In these mechanical clutches, the maximum torque threshold value may be user adjustable, often by a clutch collar that is attached to the power tool between the power tool and the tool holder/chuck. The user may rotate the clutch collar among a plurality of different positions for different maximum torque settings. The components of the mechanical clutches, however, tend to wear over time, and add excessive bulk and weight to a power tool.

(3) In order to save length and cost, some power tools additionally or alternatively include an electronic clutch. Such an electronic clutch electronically senses the output torque (e.g., via a torque transducer) or infers the output torque (e.g., by sensing another parameter such as current drawn by the motor). U.S. Pat. No. 10,220,500, which is incorporated by reference in the present patent application its entirety, provides an exemplary prior art electronic clutch. When the electronic clutch determines that the sensed output torque exceeds a threshold value, power transmission to the output shaft/spindle may be interrupted or reduced, e.g., mechanically (e.g., by actuating a solenoid to break a mechanical connection in the transmission) and/or electrically (e.g., by interrupting or reducing current delivered to the motor, and/or by actively braking the motor). Existing electronic clutches may be overly complex and/or inaccurate. For example, some electronic clutches are inaccurate because they sense current at the motor module to estimate the applied torque at the working bit. Intermediary elements between the motor and current controller (e.g., the motor & transmission) also may result in latency in sensing torque and introduce inaccuracies.

(4) Other type of clutches, such as electromagnetic clutches may feature fast activation and moderate torque density, but may require continuous electrical power to stay active. Magnetorheological clutches may be able to sense large torques, but may be heavy and may require continuous power to remain active. Because of power requirements, both of these systems may require large batteries or tethered electrical connections.

SUMMARY

(5) The present patent application provides power tools and electrostatic clutches/mechanisms for power tools.

(6) One aspect of the present patent application provides a power tool. The power tool may include a power tool housing, a motor received in the power tool housing and configured to be driven by a power source, an end effector configured to perform an operation on a workpiece, an electrostatic clutch assembly, and a control circuit. The electrostatic clutch assembly may be disposed in the power tool housing and may be between the motor and the end effector. The electrostatic clutch assembly may comprise a shaft, a first electrode and a second electrode axially arranged along an axial direction of the shaft, the first electrode and the second electrode being substantially parallel and adjacent to each other, a dielectric layer arranged between the first electrode and the second electrode; and a clutch housing containing the first electrode, the second electrode, and at least a portion of the shaft. The first electrode may be non-rotatably coupled to the clutch housing and may be axially movable relative to the clutch housing. The second electrode may be non-rotatably coupled to the shaft and may be axially movable relative to the shaft. The control circuit may be disposed in the power tool housing and may be operatively cooperable with the electrostatic clutch assembly. The control circuit causes a first voltage difference to be applied between the first electrode and the second electrode, generating a first attractive force between the first and second electrodes, causing the shaft and the clutch housing to be coupled to each other when a torque on one of the shaft or the clutch housing is less than or equal to a first threshold value, and allowing the one of the shaft or the clutch housing to rotationally slip relative to the other of the shaft or the clutch housing when the torque on the one of the shaft or the clutch housing exceeds the first threshold value.

(7) The shaft and the clutch housing may be non-rotatably coupled to each other when the torque on the shaft is less than or equal to the first threshold value. The first electrode may be one of a plurality of first electrodes in the electrostatic clutch assembly. The second electrode may be one of a plurality of second electrodes in the electrostatic clutch assembly. The first electrode and the second electrode may form an electrode pair. The electrode pair may be one of a plurality of electrode pairs that are axially arranged along the axial direction of the shaft. The plurality of electrode pairs may be substantially parallel to each other.

(8) Each first electrode of the plurality of electrode pairs may include an annular plate member, and each second electrode of the plurality of electrode pairs may include an annular plate member.

(9) Each first electrode of the plurality of electrode pairs may have a first surface and a second surface and each second electrode of the plurality of electrode pairs may have a first surface and a second surface that is facing the first surface of each first electrode. The dielectric layer may be on at least one of the first surface of each first electrode or the second surface of each second electrode. The dielectric layer may be disposed on surfaces of the first electrode and the second electrode that face each other.

(10) The electrostatic clutch assembly may further comprise a conductive material on at least one of the first surface of each first electrode or the second surface of each second electrode. The conductive material may be disposed on surfaces of each first electrode and each second electrode that face each other.

(11) Each first electrode of the plurality of electrode pairs may be electrically couplable to the power source. Each second electrode of the plurality of electrode pairs may be electrically couplable to the power source.

(12) The other of the shaft or the clutch housing may be configured to be rotationally driven by the motor.

(13) The other of the shaft or the clutch housing may be operatively connected to the motor via an input shaft that is driven by a motor assembly. The motor assembly may include the motor and a planetary transmission. The other of the shaft or the clutch housing may be non-rotatably coupled to an output carrier of the planetary transmission. The other of the shaft or the clutch housing may

be fixed to or integral with the output carrier of the planetary transmission. The other of the shaft or the clutch housing may be splined to the output carrier of the planetary transmission.

(14) The motor assembly may include the motor and a planetary transmission. The clutch housing may be mounted to a ring gear of the planetary transmission. The ring gear may be in an output stage of the planetary transmission.

(15) The electrostatic clutch assembly may be disposed between an output shaft of the motor and an input member of a planetary transmission. The other of the shaft or the clutch housing may be splined to the output shaft of the motor and the one of the shaft or the clutch housing is fixed to a first stage sun gear of the planetary transmission. The other of the shaft or the clutch housing may be fixed to or integral with the output shaft of the motor and the one of the shaft or the clutch housing is fixed to a first stage sun gear of the planetary transmission.

(16) The one of the shaft or the clutch housing of the electrostatic clutch assembly may be configured to rotationally drive the end effector that is mounted to an output spindle of the power tool. The one of the shaft or the clutch housing may be integral with or fixed to the output spindle of the power tool. The one of the shaft or the clutch housing may be splined to the output spindle of the power tool.

(17) The end effector may include a tool holder that is configured to receive a tool bit portion therein. The shaft may have a first portion and a second portion. One of the first portion and the second portion of the shaft may be operatively connected to each second electrode of the plurality of electrode pairs. The other of the first portion and the second portion of the shaft may be operatively connected to the output spindle of the power tool. The power tool may include a power drill.

(18) The end effector may be mounted to an output spindle of the power tool. The power tool may include the motor, and an impact mechanism that may be configured to apply intermittent rotational impacts to the output spindle. The impact mechanism may be a rotational impact mechanism.

(19) The impact mechanism may comprise a Pott-type impact mechanism, and wherein the other of the shaft or the clutch housing is operatively coupled to an output of the Pott-type impact mechanism. The Pott-type impact mechanism may comprise a cam shaft rotationally driven by the motor, a hammer received at least partially over the cam shaft and operatively coupled to the cam shaft to be movable axially and rotationally relative to the cam shaft, an anvil, and a spring axially biasing the hammer toward the anvil. When the torque is at or below a threshold value, the cam shaft, the hammer, and the anvil rotate in unison. When the torque exceeds the threshold value, the hammer moves axially and rotationally along the cam shaft to apply rotational impacts to the anvil. The other of the shaft or the clutch housing may be operatively coupled to the anvil.

(20) The impact mechanism may include a twin hammer impact mechanism. The other of the shaft or the clutch housing may be operatively coupled to an output of the twin hammer impact mechanism. The motor may be a pneumatic motor. The motor may be an electric motor.

(21) The impact mechanism may include an oil pulse impact mechanism. The other of the shaft or the clutch housing may be operatively coupled to an output of the oil pulse impact mechanism.

(22) The electrostatic clutch assembly may be configured to receive an input torque from the impact mechanism and produce an output torque in response thereto, the output torque being limited to a maximum threshold value and transmitted to the output spindle of the power tool. The maximum threshold value may be a fixed value. The maximum threshold value may be configured to be adjusted dynamically. The maximum threshold value may be a variable value. The maximum threshold value may be a function of time.

(23) Another aspect of the present patent application provides a power tool. The power tool may comprise a power tool housing, a motor received in the power tool housing and configured to be driven by a power source, an output spindle at least partially received in and rotatable relative to the power tool housing, an impact mechanism, an electrostatic clutch assembly, and a control circuit. The impact mechanism may be operatively coupled with the motor and may be configured

to be driven thereby. The impact mechanism may be configured to generate intermittent rotational impacts that are transmitted to the output spindle of the power tool. The electrostatic clutch assembly may be disposed in the power tool housing and may be between the impact mechanism and the output spindle. The electrostatic clutch assembly may comprise a shaft, a first electrode and a second electrode axially arranged along an axial direction of the shaft, the first electrode and the second electrode being substantially parallel to each other, and a clutch housing containing the first electrode, the second electrode and at least a portion of the shaft. The first electrode may be non-rotatably coupled to the clutch housing and is axially movable relative to the clutch housing. The second electrode may be non-rotatably coupled to the shaft and is axially movable relative to the shaft. The control circuit may be disposed in the power tool housing and may be operatively cooperable with the electrostatic clutch assembly. The control circuit causes a first voltage difference to be applied between the first electrode and the second electrode, generating a first attractive force between each of the first and second electrodes, causing the shaft to rotate together with the clutch housing when a torque on the shaft is less than or equal to a first threshold value and causing the shaft to rotationally slip relative to the clutch housing when the torque on the shaft exceeds the first threshold value. The electrostatic clutch assembly is configured to receive an input torque from the impact mechanism and produce an output torque in response thereto, the output torque being limited to a maximum threshold value and transmitted to the output spindle of the power tool.

(24) The maximum threshold value may be a fixed value. The maximum threshold value may be configured to be adjusted dynamically. The maximum threshold value may be a variable value. The maximum threshold value may be a function of time. The input torque from the impact mechanism may vary in magnitude and/or duration. The output torque from the impact mechanism may vary in duration and may be constant in magnitude.

(25) The impact mechanism may be a rotational impact mechanism.

(26) The impact mechanism may comprise a Pott-type impact mechanism that includes a cam shaft rotationally driven by the motor, a hammer received at least partially over the cam shaft and operatively coupled to the cam shaft to be movable axially and rotationally relative to the cam shaft, an anvil, and a spring axially biasing the hammer toward the anvil. One of the shaft or the clutch housing is operatively coupled to an output of the Pott-type impact mechanism.

(27) The impact mechanism may comprises a Pott-type impact mechanism that includes a cam shaft rotationally driven by the motor, a hammer received at least partially over the cam shaft and operatively coupled to the cam shaft to be movable axially and rotationally relative to the cam shaft, an anvil, and a spring axially biasing the hammer toward the anvil. One of the shaft or the clutch housing is operatively coupled to the anvil.

(28) The impact mechanism may include a twin hammer impact mechanism. One of the shaft or the clutch housing is operatively coupled to an output of the twin hammer impact mechanism.

(29) The motor may be a pneumatic motor. The motor may be an electric motor.

(30) The impact mechanism may include an oil pulse impact mechanism. One of the shaft or the clutch housing is operatively coupled to an output of the oil pulse impact mechanism.

(31) The electrostatic clutch assembly may be configured to receive an input torque from the impact mechanism and produce an output torque in response thereto. The output torque may be limited to a maximum threshold value and transmitted to the output spindle of the power tool.

(32) Yet another aspect of the present patent application provides a power tool. The power tool may comprise a power tool housing, a motor received in the power tool housing and configured to be driven by a power source, a planetary transmission configured to be rotationally driven by the motor, an end effector, an electrostatic clutch assembly, and a control circuit. The planetary transmission may include an output planet carrier as a rotational output of the transmission. The end effector may be configured to perform an operation on a workpiece. The electrostatic clutch assembly may be disposed in the power tool housing between the planetary transmission and the

end effector. The electrostatic clutch assembly may include an input member non-rotatably coupled to a first electrode, an output member non-rotatably coupled to a second electrode, and a dielectric layer arranged between the first and second electrodes. The control circuit may be disposed in the power tool housing and may be operatively cooperable with the electrostatic clutch assembly to control power delivery from the power source to the first electrode and the second electrode. The input member may be operatively coupled to the output planet carrier of the transmission to rotate with the output planet carrier and the output member may be operatively coupled to the end effector to rotate with the end effector. The control circuit causes a first voltage difference to be applied between the first electrode and the second electrode, generating a first attractive force between the first and second electrodes, causing the output member and the input member to be coupled to each other when a torque on the output member is less than or equal to a first threshold value, and allowing the output member to rotationally slip relative to the input member when the torque on the output member exceeds the first threshold value.

(33) The input member of the electrostatic clutch assembly may be fixed to or integral with the output carrier of the planetary transmission. The input member of the electrostatic clutch assembly may be splined to the output carrier of the planetary transmission.

(34) The input member of the electrostatic clutch assembly may be coupled to a ring gear of the planetary transmission.

(35) The ring gear may be in an output stage of the planetary transmission.

(36) The electrostatic clutch assembly may be disposed between an output shaft of the motor and an input member of the planetary transmission.

(37) One of the shaft or the clutch housing may be fixed to a first stage sun gear of the planetary transmission and the other of the shaft or the clutch housing may be splined to the output shaft of the motor and the one of the shaft or the clutch housing is fixed to a first stage sun gear of the planetary transmission.

(38) One of the shaft or the clutch housing may be fixed to a first stage sun gear of the planetary transmission and the other of the shaft or the clutch housing may be fixed to or integral with the output shaft of the motor and the one of the shaft or the clutch housing is fixed to a first stage sun gear of the planetary transmission.

(39) Yet another aspect of the present patent application provides a power tool. The power tool may comprise a power tool housing, a motor received in the power tool housing and configured to be driven by a power source, an end effector configured to perform an operation on a workpiece, a planetary transmission configured to be rotationally driven by the motor, an electrostatic clutch assembly, and a control circuit. The planetary transmission may include a ring gear. The electrostatic clutch assembly may be disposed in the power tool housing between the planetary transmission and the end effector. The electrostatic clutch assembly may include a first member non-rotatably coupled to a first electrode, a second member non-rotatably coupled to a second electrode, and a dielectric layer arranged between the first and second electrodes. The control circuit may be disposed in the power tool housing and may be operatively cooperable with the electrostatic clutch assembly to control power delivery from the power source to the first electrode and the second electrode. The first member may be non-rotatably coupled to the ring gear and the second member may be non-rotatably coupled to the power tool housing. The control circuit causes a first voltage difference to be applied between the first electrode and the second electrode, generating a first attractive force between the first and second electrodes, causing the ring gear to remain stationary relative to the power tool housing, enabling torque transmission from the planetary transmission to the end effector, when a torque on the second member is less than or equal to a first threshold value, and allowing the ring gear to rotate relative to the power tool housing, interrupting torque transmission from the planetary transmission to the end effector, when the torque on the second member exceeds the first threshold value. The ring gear may be in an output stage of the planetary transmission.

(40) Yet another aspect of the present patent application provides a power tool. The power tool may comprise a power source, a motor configured to be driven by the power source, an end effector configured to perform an operation on a workpiece, an electrostatic clutch assembly between the motor and the end effector, a control circuit, and a switch circuit. The electrostatic clutch assembly may comprise a first electrode and a second electrode that is substantially parallel and adjacent to each other; and a dielectric layer arranged between the first electrode and the second electrode. The control circuit may be operatively cooperable with the electrostatic clutch assembly to control power delivery from the power source to the first electrode and the second electrode. The control circuit causes a first voltage difference to be applied between the first electrode and the second electrode. The switch circuit may be operatively cooperable with the control circuit. The switch circuit may be configured to selectively apply power to the first electrode and the second electrode. The switch circuit may also be configured to change a voltage polarity of the first electrode and the second electrode, such that in a first configuration, the first electrode is at a first voltage at a first polarity and the second electrode is at a second voltage at a second polarity, and in a second configuration, the first electrode is at the first voltage at the second polarity and the second electrode is at the second voltage at the first polarity.

(41) The switch circuit may further be configured to drain residual voltages from the first and second electrodes. The power tool may further comprise a sense circuit that is configured to sense a voltage potential measurement representative of a voltage difference between the first voltage applied to the first electrode and the second voltage applied to the second electrode. The sense circuit may be configured to provide the voltage potential measurement as an input to the control circuit. The control circuit may be configured to determine, based on the input, the first voltage to be applied to the first electrode and the second voltage to be applied to the second electrode.

(42) The sense circuit may be configured to sense a current measurement representative of a current flow between the first electrode and the second electrode. The sense circuit may be configured to provide the current measurement as an input to the control circuit. The control circuit may be configured to determine, based on the input, the first voltage to be applied to the first electrode and the second voltage to be applied to the second electrode.

(43) The sense circuit may be configured to sense a motor rotation speed measurement. The sense circuit may be configured to provide the motor rotation speed measurement as an input to the control circuit. The control circuit may be configured to determine, based on the input, the first voltage to be applied to the first electrode and the second voltage to be applied to the second voltage.

(44) The sense circuit may be configured to sense a reaction torque measurement. The sense circuit may be configured to provide the reaction torque measurement as an input to the control circuit. The control circuit may be configured to determine an error using the reaction torque measurement and an applied torque. The control circuit may be configured to determine, based on the error, the first voltage to be applied to the first electrode and the second voltage to be applied to the second electrode.

(45) The sense circuit may be configured to sense a torque setting interface measurement. The sense circuit may be configured to provide the torque setting interface measurement as an input to the control circuit. The control circuit may be configured to determine, based on the input, the first voltage to be applied to the first electrode and the second voltage to be applied to the second electrode.

(46) Yet another aspect of the present patent application provides a method of operating a power tool. The power tool may comprise a motor, an electrostatic clutch assembly, an output spindle, and a control circuit. The electrostatic clutch assembly may comprise a first electrode, a second electrode and a dielectric layer arranged between the first electrode and the second electrode. The electrostatic clutch assembly may be positioned between the motor and the output spindle. The method of operating a power tool comprises actuating the motor; applying a first voltage difference

between the first electrode and the second electrode; controlling the motor to increase a torque applied to the output spindle until the torque applied to the output spindle reaches a first threshold value. One of the first electrode and the second electrode rotationally slips relative to the other of the first electrode and the second electrode when the torque on the output spindle exceeds the first threshold value. The method of operating a power tool may also comprise initiating a protective action upon detecting that one of the first electrode and the second electrode rotationally slips relative to the other of the first electrode and the second electrode.

(47) The initiating a protective action may include discharging the first voltage from the first electrode and the second voltage from the second electrode. The initiating a protective action may include reducing or interrupting power delivery to the motor. The initiating a protective action may include braking the motor. The first threshold value may correspond to a holding force between the first electrode and a second electrode. The holding force may correspond to a magnitude of a difference between the first voltage and the second voltage. The holding force is proportional to a square of the magnitude of the difference between the first voltage and the second voltage.

(48) These and other aspects of the present patent application, as well as the methods of operation and functions of the related elements of structure and the combination of parts and economies of manufacture, will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. In one embodiment of the present patent application, the structural components illustrated herein are drawn to scale. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the present patent application. It shall also be appreciated that the features of one embodiment disclosed herein can be used in other embodiments disclosed herein. As used in the specification and in the claims, the singular form of “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows an exemplary power tool according to various embodiments of the present patent application;
- (2) FIG. 2 shows a cross-sectional side view of a portion of the power tool of FIG. 1, the power tool having an exemplary electrostatic clutch assembly according to various embodiments of the present patent application;
- (3) FIGS. 3 and 4 show cross-sectional views of an electrostatic clutch assembly according to various embodiments of the present patent application;
- (4) FIG. 5 shows an exploded view of the electrostatic clutch assembly of FIGS. 3 and 4 according to various embodiments of the present patent application;
- (5) FIGS. 5A and 5B show additional cross-sectional views of the electrostatic clutch assembly of FIGS. 3 and 4, where some portions of the electrostatic clutch assembly are not shown in FIGS. 3-5B to better illustrate other portions of the electrostatic clutch assembly;
- (6) FIG. 6 shows another cross-sectional view of the exemplary power tool of FIG. 1, the power tool having an electrostatic clutch assembly according to various embodiments of the present patent application, where the electrostatic clutch assembly is configured to be integrally mounted to an output carrier of a planetary transmission;
- (7) FIGS. 7-8 show partial exploded views of the power tool of FIG. 6, the power tool having the electrostatic clutch assembly according to various embodiments of the present patent application, where some portions of the power tool are not shown to better illustrate other portions of the power

tool;

(8) FIGS. **9A-9B** show the electrostatic clutch assembly according to the embodiment of FIG. **6**, where the electrostatic clutch assembly is configured to be integrally mounted to the output carrier of the planetary transmission;

(9) FIGS. **10** and **11** show a partial assembled view and a partial exploded view, respectively, of another exemplary power tool having an electrostatic clutch assembly according to various embodiments of the present patent application, where some portions of the power tool are not shown to better illustrate other portions of the power tool, and where the electrostatic clutch assembly is configured to be splined to the output carrier of the planetary transmission;

(10) FIG. **12** shows a cross-sectional view of another exemplary embodiment of a power tool having an electrostatic clutch assembly according to various embodiments of the present patent application, where the electrostatic clutch is positioned between a motor and a transmission of the power tool;

(11) FIGS. **13** and **14** show a perspective view and a cross-sectional view, respectively, of another exemplary embodiment of a power tool with an electrostatic clutch assembly according to various embodiments of the present patent application, where the electrostatic clutch assembly is configured to be coupled to a ring gear of a transmission;

(12) FIG. **15** shows a perspective view of an exemplary embodiment of a rotary impact power tool with a Pott-type impact mechanism and an electrostatic clutch assembly according to various embodiments of the present patent application;

(13) FIGS. **16** and **17** show a partial exploded view and a cross-sectional view, respectively, of the impact tool of FIG. **15**;

(14) FIG. **18** shows a perspective view of another exemplary embodiment of a rotary impact power tool with a twin hammer impact mechanism and an electrostatic clutch assembly according to various embodiments of the present patent application;

(15) FIG. **19** shows a cross-sectional view of the impact tool of FIG. **18**;

(16) FIG. **20** shows a cross-sectional view of another exemplary embodiment of a rotary impact power tool with a hydraulic impulse impact mechanism and an electrostatic clutch assembly according to various embodiments of the present patent application;

(17) FIG. **21** shows a schematic block diagram of an exemplary power tool control system with an electrostatic clutch assembly according to various embodiments of the present patent application;

(18) FIG. **22** shows an exemplary power tool with an electrostatic clutch assembly according to various embodiment of the present application, shown in use with an applied torque and a reaction torque being used to calculate an error;

(19) FIG. **23** show an exemplary torque setting user interface that converts numerical setpoints to a clutch setting voltage, which is provided as an input to a controller of a power tool according to various embodiments of the present patent application;

(20) FIG. **24** shows an exemplary graphical representation of an operation of an exemplary power tool with an exemplary electrostatic clutch according to various embodiments of the present patent application, where time is shown on the x-axis and applied torque and motor speed are shown on the y-axis;

(21) FIGS. **25A-26B** show exemplary graphical representations of operation of exemplary power tool with an exemplary electrostatic clutch assembly, where the power tool includes a rotary impact mechanism;

(22) FIG. **27** shows a graphical representation of an operation of an electrostatic clutch assembly in a torque-limit mode according to various embodiments of the present patent application;

(23) FIGS. **28A** and **28B** show exemplary power tools with exemplary indicators of slip indicating achieved torque setting according to various embodiments of the present patent application;

(24) FIG. **29** shows a graphical representation of an operation of an electrostatic clutch assembly in a power tool in which alternating polarity is provided in each cycle according to various

embodiments of the present patent application;

(25) FIG. 30 shows a graphical representation of an operation of an electrostatic clutch assembly in a power tool according to various embodiments of the present patent application, where short intervals of slip of the electrostatic clutch assembly followed by re-engagement is shown so as to prevent a fastener bit of the power tool from camming out or stripping a fastener head; and

(26) FIG. 31 shows a graphical representation of an operation of an electrostatic clutch assembly in an impact power tool according to various embodiments of the present patent application, where short intervals of slip of the electrostatic clutch assembly followed by re-engagement is shown to simulate operation of an impact mechanism in a direct drive power tool.

DETAILED DESCRIPTION

(27) In an embodiment, the present patent application provides electrostatic clutches or electroadhesive clutches for power tools.

(28) Referring to FIGS. 1-2, 6-8, and 10-20, the present patent application provides a power tool 10. As a person of ordinary skill in the art would appreciate, embodiments may include either a corded power tool/device or a cordless (battery operated) power tool/device. In one embodiment, the power tool 10 is a portable device.

(29) In one embodiment, the power tool 10 is a power drill or drill/driver. The power tool 10 also may be a power screwdriver, a power fastener/fastening tool, a power driver, a power expansion tool, and/or other power devices/tools. An exemplary power drill 10 is shown in FIGS. 1-2, 6-8, and 10-14. In another embodiment, the power tool may be an impact tool/device. The power tool may be a power rotary impact tool, a power impact tool, a power impact driver, a power impact wrench, and/or other power devices/tools. An exemplary power impact tool is shown in FIGS. 15-20.

(30) The power tool 10 includes a housing 12, a motor and transmission assembly 14, an end effector 22, an electrostatic clutch assembly 18, and a control circuit 50. The housing 12 may be interchangeably referred to as power tool housing. The housing 12 may have a power supply interface 102 configured to be coupled to a power source or an electrical power source 2, such as a battery pack 26. The motor and transmission assembly 14 is received in the housing 12. In one embodiment, the motor and transmission assembly 14 may include a motor 15 and a transmission 16, such as a multi-speed transmission assembly, that are received in the housing 12. In another embodiment, the motor and transmission assembly 14 may include the motor 15 that is received in the housing 12. The motor 15 is configured to be powered by the power source 26.

(31) Referring to FIG. 1, the housing 12 may include a pair of mating handle shells 34 that cooperate to define a handle portion 36 and a drive train or body portion 38. In one embodiment, the body portion 38 includes a motor receiving portion and a transmission receiving portion. The handle portion 36 is configured to be coupled to the power source. In one embodiment, the power source includes a battery pack 26 or an AC power source (not shown) as described in detail below.

(32) The power tool 10 may also include a trigger assembly 24. The trigger assembly 24 and the battery pack 26 are mechanically coupled to the handle portion 36 and are electrically coupled to the motor and transmission assembly 14 in a conventional manner that is not specifically shown but which is readily understood by and within the capabilities of a person of ordinary skill in the art. In one embodiment, the power tool 10 includes other sources of power (e.g., alternating current (AC) power cord/cable, compressed air source and/or other sources of power) coupled to a distal end of the handle portion 36. In one embodiment, the trigger assembly 24 is a variable speed trigger. The trigger assembly 24 may be configured to be operatively coupled to the housing 12 for selectively actuating and controlling the speed of the motor 15, for example, by controlling a pulse width modulation (PWM) signal delivered to the motor 15.

(33) The power tool 10 may also include an output member/spindle 68 that is at least partially received in and rotatable relative to the power tool housing 12. The output member/spindle 68 may be interchangeably referred to as output shaft 68.

(34) The end effector **22** of the power tool **10** is mounted to the output spindle **68** of the power tool **10**. The end effector **22** is operatively coupled to the housing **12** and is configured to perform an operation on a workpiece (not shown). As shown in FIG. 2, the end effector **22** may include or may be interchangeably referred to as chuck/tool holder **22**. The tool holder **22** may be configured for holding a power tool accessory, such as a drill bit, an expansion bit, a screwdriver bit, and/or other tool bits. The tool holder **22** may include a tool bit holder portion, such as chuck jaws **23**, therein. In one embodiment, the tool holder **22** may be a keyless chuck, although it should be understood that the tool holder can have other tool holder configurations such as a quick release tool holder, a hex tool holder, or a keyed tool holder/chuck.

(35) A person of ordinary skill in the art would understand that several of the components of the power tool **10**, such as the tool holder **22**, the trigger assembly **24** and the battery pack **26**, are conventional in nature and therefore need not be discussed in significant detail in the present patent application. Reference may be made to a variety of patents/patent publications for a more complete understanding of the conventional features of the power tool **10**. Examples of such patents include U.S. Pat. No. 5,897,454 issued Apr. 27, 1999 and U.S. Pat. No. 6,431,289, issued Aug. 13, 2002, which are hereby incorporated by reference in their entirety in the present patent application.

(36) The power tool or impact tool **10** may include an impact mechanism **402** that is operatively coupled to the motor **15** (e.g., via the transmission **16**) and is configured to be driven thereby. The impact mechanism **402** is configured to generate intermittent rotational impacts that are transmitted to the output spindle **68** of the power tool **10**. The impact mechanism **402** will be described in detail below with respect to FIGS. 15-20.

(37) In one embodiment, the motor **15** is disposed/received in the housing **12**. The motor **15** may be housed in the motor receiving portion of the drive train or body portion **38** of the housing **12**. The motor **15** may include a rotatable output motor shaft, which extends into the transmission receiving portion of the drive train or body portion **38** of the housing **12**. A motor pinion having a plurality of gear teeth may be coupled for rotation with the rotatable output motor shaft. The motor **15** may be a brushless or electronically commutated motor, although the motor **15** may be another type of brushed DC motor or universal motor. The trigger assembly **24** and battery pack **26** may cooperate to selectively provide electric power to the motor and transmission assembly **14** in a manner that is generally well known in the art so as to permit the user of the power tool **10** to control the speed and direction with which the rotatable output motor shaft rotates.

(38) The motor output shaft extends from the motor **15** to the transmission **16**, which transmits torque from the motor output shaft to one of a shaft **302** and a clutch housing **310** of the electrostatic clutch assembly **18**, which transmits torque to the other of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** and to the end effector **22**. The electrostatic clutch assembly **18** and the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** are all described in detail below. The motor **15** is configured to provide a torque to one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** (e.g., via the transmission **16**).

(39) The transmission **16** may be housed in the transmission receiving portion of the drive train or body portion **38** of the housing **12**. The transmission **16** may comprise a single speed transmission. The transmission **16** may comprise a multi-speed transmission having a plurality of gears and settings that allow the speed reduction through the transmission **16** to be changed, in a manner well understood to a person of ordinary skill in the art. The transmission **16** may include a multi-stage planetary gear set, with each stage having an input sun gear, a plurality of planet gears meshed with the sun gear and pinned to a rotatable planet carrier, and a ring gear meshed with and surrounding the planet gears. For each stage of the multi-stage planetary gear set, if a ring gear is rotationally fixed relative to the housing **12**, the planet gears orbit the sun gear when the sun gear rotates, transferring power at a reduced speed to their planet carrier, thus causing a speed reduction through that stage. If a ring gear is allowed to rotate relative to the housing **12**, then the sun gear causes the

planet carrier to rotate at the same speed as the sun gear, causing no speed reduction through that stage. By varying which one or ones of the stages have the ring gears are fixed against rotation, the total amount of speed reduction through the transmission **16** can be controlled, and thus the speed setting of the transmission **16** (e.g., among high, medium, and low) can be adjusted. This adjustment of the speed setting is achieved via a shift ring that surrounds the ring gears and that is shiftable along an axis of the output shaft to lock different stages of the ring gears against rotation. The power tool **10** may include a speed selector switch for selecting the speed reduction setting of the transmission **16**. The speed selector switch is operatively coupled to the shift ring by spring biased pins so that axial movement of the speed selector switch causes the axial movement of the shift ring. Further details regarding an exemplary multi-speed transmission is described in U.S. Pat. No. 7,452,304 which is incorporated by reference in its entirety in the present patent application. It should be understood that other types of single speed or multi-speed transmissions and other mechanisms for shifting the transmission among the speeds are within the scope of the present patent application. For example, the transmission may include a parallel axis transmission, a spur gear transmission, a right angle transmission. In yet further embodiments, there may be no transmission and the motor may directly drive the input shaft or clutch housing of the electrostatic clutch assembly.

(40) In various embodiments, the electrostatic clutch assembly **18** of the present patent application may employ features of the electrostatic clutches disclosed in detail in U.S. Pat. Nos. 10,663,016; 10,138,953; and/or Japanese Utility Model Publication No. Sho 48-41469, each of which is incorporated by reference in their entirety in the present patent application. In another embodiment, the electrostatic clutch assembly **18** may include structure and operation as described in detail with respect to FIGS. 3-5B of the present patent application.

(41) The electrostatic clutch assembly **18** may be disposed in the power tool housing **12**. Referring to FIGS. 6-11 and 13-14, in some embodiments, the electrostatic clutch assembly **18** may be disposed between the transmission assembly **14** and the end effector **22**.

(42) Referring to FIG. 12, in another embodiment, the electrostatic clutch assembly **18** may be disposed between the transmission **16** and the motor **15**.

(43) Referring to FIGS. 15-20, in yet other embodiments, the electrostatic clutch assembly **18** may be disposed in the power impact tool housing **12** between an output spindle/shaft **68** and an impact mechanism of an impact power tool **10**.

(44) Referring to FIGS. 3-5B, the electrostatic clutch assembly **18** comprises a shaft **302**, a first electrode **304**, a second electrode **306**, a dielectric layer **308**, and a clutch housing **310**. The first electrode **304** and the second electrode **306** are axially arranged along an axial direction (e.g., AD-AD in FIG. 3) of the shaft **302** and/or the clutch housing **310**.

(45) The first electrode **304** and the second electrode **306** are substantially parallel and are disposed adjacent to each other. The first and second electrodes **304**, **306** are generally planar. The first and second electrodes **304**, **306** may generally have annular plate/disc shaped configurations.

(46) One of the first electrode **304** and the second electrode **306** is mounted on the shaft **302**. For example, in the illustrated embodiment in FIGS. 3-5B, the electrode **306** may be referred to as a shaft electrode. The shaft electrode **306** is mounted on the shaft **302** such that the second electrode **306** and the shaft **302** rotate together. That is, the shaft electrode **306** does not touch the clutch housing **310**.

(47) Referring to FIG. 5, the shaft electrode **306** includes an annular plate/disc shaped configuration having outer circumference **3060** and an inner circumference **3061**. The inner circumference **3061** of the shaft electrode **306** may include a toothed arrangement having teeth **306T** that are configured to be received in grooves **302R** (e.g., on an outer circumferential surface) on the shaft **302** when the second electrode **306** is mounted on the shaft **302**. The second electrode **306** may also include an opening **347** that is configured to receive the shaft **302** therein.

(48) The other of the first electrode **304** and the second electrode **306** is arranged along the shaft

302. The other of the first electrode **304** and the second electrode **306** may or may not touch the shaft **302**. For example, in the illustrated embodiment in FIGS. 3-5B, the electrode **304** may be referred to as a housing electrode. The housing electrode **304** is mounted to the clutch housing **310** such that housing electrode **304** and the clutch housing **310** rotate together. That is, the housing electrode **304** does not touch the shaft **302**.

(49) The housing electrode **304** includes an annular plate/disc shaped configuration having outer circumference **3040** and an inner circumference **3041**. The outer circumference **3040** of the housing electrode **304** has a toothed arrangement having teeth **304T** that are configured to be received in grooves **310R** on the inner (circumferential) surface **333** of the clutch housing **310** when the first electrode **304** is received in the clutch housing **310**. The first electrode **304** may also include an opening **345** that is configured to receive the shaft **302** therein.

(50) One of the first electrode **304** and the second electrode **306** may also be referred to as an input member/electrode of the electrostatic clutch assembly **18** that is configured to be rotationally driven by the motor and transmission assembly **14** or driven by the impact mechanism **402**. That is, one of the first electrode **304** and the second electrode **306** along with the shaft **302** or clutch housing **310** to which it is non-rotatably coupled are configured to be rotationally driven by the motor and transmission assembly **14** or driven by the impact mechanism **402**.

(51) The other of the first electrode **304** and the second electrode **306** may also be referred to as an output member/electrode of the electrostatic clutch assembly **18** that is configured to rotationally drive the end effector **22**. That is, the other of the first electrode **304** and the second electrode **306** along with the shaft **302** or clutch housing **310** to which it is non-rotatably coupled are configured to rotationally drive the end effector **22** via the output shaft/spindle **68**.

(52) The first electrode **304** is one of a plurality of first electrodes **304** in the electrostatic clutch assembly **18** and the second electrode **306** is one of a plurality of second electrodes **306** in the electrostatic clutch assembly **18**. The first electrode **304** and the second electrode **306** form an electrode pair **307**. The electrode pair **307** is one of a plurality of electrode pairs **307** that are axially arranged along the axial direction AD-AD of the shaft **302**, and the plurality of electrode pairs **307** is substantially parallel to each other. The number of electrode pairs **307** (i.e., each including the first electrode **304** and the second electrode **306**) may vary and may depend on the application.

(53) As discussed above, each first electrode **304** of the plurality of electrode pairs **307** includes an annular plate member/configuration, and each second electrode **306** of the plurality of electrode pairs **307** includes an annular plate member/configuration.

(54) In the illustrated embodiment of FIGS. 3-5B, each first electrode **304** is non-rotatably coupled to the clutch housing **310** (so that it will rotate with the clutch housing) and each second electrode **306** is non-rotatably coupled to the shaft **302** (so that it will rotate with the shaft **302**).

(55) The term axially movable may include an axial movement between two components. At least one of the electrodes **304**, **306** is axially moveable relative to the other electrode **304**, **306**. In one implementation, the first electrode **304** is axially movable (i.e., along the axial direction (e.g., AD-AD in FIG. 3) of the shaft **302**/the clutch housing **310**) relative to the clutch housing **310**, while the second electrode **306** is axially fixed to the shaft **302**. In another implementation, the second electrode **306** is axially movable (i.e., along the axial direction (e.g., AD-AD in FIG. 3) of the shaft **302**/the clutch housing **310**) relative to the shaft **302**, while the first electrode is axially fixed to the clutch housing **310**. In yet another implementation, both the first electrode **304** is axially movable relative to the clutch housing **310** and second electrode **306** is axially movable relative to the shaft **302**. In yet further implementations, the first electrode **304** may be axially fixed to the clutch housing **310**, the second electrode **306** may be axially fixed to the shaft **302**, and one or both of the clutch housing **310** and the shaft **302** may be axially moveable relative to the other of the clutch housing **310** and the shaft **302**.

(56) The term non-rotatably coupled may include two components that rotate together. As each first electrode **304** is non-rotatably coupled to the clutch housing **310**, each first electrode **304** and the

clutch housing **310** rotate together. Similarly, as each second electrode **306** is non-rotatably coupled to the shaft **302**, each second electrode **306** and the shaft **302** rotate together.

(57) International Patent Application No.: PCT/US2022/022346 (“PCT Application '346”), filed Mar. 29, 2022, titled “ELECTROSTATIC CLUTCH FOR POWER TOOL,” which is hereby incorporated by reference in its entirety. As is disclosed in detail and is claimed in the PCT application '346, in an alternative embodiment, the electrostatic clutch assembly **18** may include a single electrode (e.g., one of the electrode **304** or **306**) coupled to an input shaft and a single electrode (e.g., the other of the electrode **304** or **306**) coupled to an output shaft, with no clutch housing, and at least one electrode **304**, **306** axially moveable relative to the other electrode **304**, **306**.

(58) The first electrode **304** and the second electrode **306** are electrically couplable to the power source **102**. In FIG. 3, only one first electrode **304** of the plurality of electrode pairs **307** and only one second electrode **306** of the plurality of electrode pairs **307** are shown electrically couplable to the power source **102** for sake of clarity and ease of understanding. A person of ordinary skill in the art would appreciate that each first electrode **304** of the plurality of electrode pairs **307** is electrically couplable to the power source **102** and each second electrode **306** of the plurality of electrode pairs **307** is electrically couplable to the power source **102**.

(59) In another embodiment, each first electrode **304** (and the clutch housing **310**) and each second electrode **306** (and the shaft **302**) are electrically couplable to the electrical power source **102** via brushes **371**, springs **373**, discs **363**, **367**, inner brush **365**, etc. as described in detail below with respect to FIGS. 5A-5B.

(60) The dielectric layer **308** is arranged between each first electrode **304** and each second electrode **306**. The dielectric layer **308** separates each first electrode **304** and each second electrode **306**. With a pair **307** of electrodes **304**, **306**, at least one electrode **304**, **306** is covered in the dielectric material/layer **308** to maintain the gap between the conductive surfaces of the electrodes **304**, **306**. The two electrostatic films are always separated by the dielectric layer **308**. The dielectric layer is very thin and has dimension in the order of microns. The dielectric layer **308** is a layer of dielectric material. If the dielectric layer is a layer of material **308**, the dielectric layer of material **308** is layered over one or both of the electrodes **304**, **306**.

(61) When the electrodes **304**, **306** are energized, the dielectric layer **308** of one electrode **304**, **306** is configured to touch the other electrode **304**, **306**. When the electrodes **304**, **306** are energized, the dielectric layer **308** of one electrode **304**, **306** is configured to touch the dielectric layer **308** of the other electrode **304**, **306**. When the electrodes **304**, **306** are deenergized, there may be an air gap between the dielectric layer **308** on one electrode **304**, **306** and the other electrode **304**, **306**. When the electrodes **304**, **306** are deenergized, there may be an air gap between the dielectric layer **308** on one electrode **304**, **306** and the dielectric layer **308** on the other electrode **304**, **306**. In one embodiment, it is not necessary for both electrodes **304**, **306** to have a dielectric layer **308** and the electrostatic clutch assembly **18** works just fine, and maybe better, with a dielectric layer **308** on only one electrode **304**, **306**.

(62) Referring to FIG. 3, each first electrode **304** of the plurality of electrode pairs **307** has a first surface **304F** and a second surface **304S** and each second electrode **306** of the plurality of electrode pairs **307** has a first surface **306F** and a second surface **306S** that is facing the first surface **304F** of each first electrode **304**. The dielectric layer **308** is on at least one of the first surface **304F** of each first electrode **304** or the second surface **306S** of each second electrode **306**. The dielectric layer **308** is disposed on surface **304F** of the first electrode **304** and surface **306S** the second electrode **306** that face each other.

(63) Each of the first electrode **304** and the second electrode **306** may include a substrate and an electrostatic layer/coating/film deposited thereon. Each of the facing surfaces **304F**, **306S** of the first electrode **304** and the second electrode **306** are coated with the electrostatic layer. In one embodiment, the electrostatic clutch assembly **18** may include a plurality of layers of electrostatic

layers that are layered on top of one another for greater holding force.

(64) Each of the first electrode **304** and the second electrode **306** may include a conductive material disposed on at least a portion thereof. In one embodiment, the conductive material is disposed beneath surfaces **304F**, **306S** of the first electrode **304** and the second electrode **306** that face each other. Each of the first electrode **304** and the second electrode **306** includes at least one conductive material/layer. The conductive material is disposed on surfaces **304F**, **306S** of each first electrode **304** and each second electrode **306** that face each other. The conductive material on at least one of the first surface **304F** of each first electrode **304** or the second surface **306S** of each second electrode **306**.

(65) Each of the first and second electrodes **304**, **306** and/or the electrostatic layer/coating/film comprise a lightweight conductive material, such as aluminum-sputtered biaxially-oriented polyethylene terephthalate. Each of the first and second electrodes **304**, **306** may include aluminum-sputtered BOPET (Bi-axially Oriented Polyethylene Terephthalate) film, also known as Mylar® film. The aluminum deposition acts as the conductive layer and the BOPET acts as the substrate. Aluminum-sputtered BOPET films of this type can have a thickness of around 25 microns. Despite the thin profile, the material is sufficiently strong to act as a force transmission component. In addition, very little electrode material is required to hold a charge, making thin and lightweight electrodes **304**, **306** possible. In alternative embodiments, a single-layer, conductive electrode, such as a metallic foil, is used.

(66) The electrostatic clutch assembly **18** may include a plurality of clutch settings. Each clutch setting may correspond to a desired output operation of the power tool **10**. That is, the clutch setting of the electrostatic clutch **18** can be set by the user based on a desired output operation. For example, the desired output operation can include an amount of material to be removed from a workpiece. Each clutch setting may have the set or predetermined torque. Each clutch setting may be associated with a different clutch disengage torque (i.e., a torque at which the electrostatic clutch assembly **18** disengages to thereby prevent the transmission of torque transmission between the motor and transmission assembly **14** or the impact mechanism **402** and the output shaft **68**). Each predetermined clutch setting may include a maximum clutch setting, a minimum clutch setting, and a plurality of intermediate clutch settings between the maximum and minimum clutch settings.

(67) Each predetermined clutch setting may at least include a first configuration in which the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** are non-rotatably coupled to each other when a torque on the output spindle **68** is less than or equal to the predetermined clutch setting, and a second configuration in which the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** rotationally slip relative to each other when the torque on the output spindle **68** is greater than the predetermined clutch setting. The power tool **10** may also include a clutch setting switch or collar that may be used by a user so as to adjust a clutch setting of the electrostatic clutch assembly **18**.

(68) When a voltage above a certain minimum threshold is applied to the electrodes **304**, **306**, the electrodes **304**, **306** are charged and electrostatically attract with an attractive force to non-rotatably couple to each other, such that the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** rotate together. Below the minimum threshold of the applied voltage, the electrodes **304**, **306** are allowed to fully decouple so that one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** can rotate relative to the other of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18**. Above the minimum threshold of voltage, the amount of holding force between the electrodes **304**, **306** corresponds to the amount of voltage applied to the electrodes, such that the holding force increases when the voltage increases above the minimum threshold. In an implementation, the holding force is proportional to the square of the voltage applied to the electrodes. When a torque applied to the shaft **302** or the housing **310** exceeds a threshold amount that corresponds to the holding force between the electrodes **304**, **306**, the electrodes **304**, **306** may slip or rotate relative to one another, which interrupts torque

transmission between the shaft and the clutch housing. In one embodiment, the transitions between the states of the clutch being fully engaged and slipping may have an observable electrical signature/signal or may be sensed by another sensor, such as a Hall sensor, and may be transmitted to the power tool control module(s)/controller/control circuit **50**, such that a response to this may be ceasing rotation of the power tool's motor **15** and/or discharging the electrostatic charge from the electrodes. In one embodiment, the torque transfer function may be descriptively a function of charge rather than voltage.

(69) In one embodiment, depending on the amount of voltage applied to the electrodes **304**, **306**, the charge and their holding force can vary. There can be different clutch settings for different amounts of voltage applied to the electrodes **304**, **306**. The applied voltage corresponding to the torque at which the electrostatic clutch assembly **18** begins to slip or rotate is different for different clutch settings (e.g., greater applied voltage corresponds to greater torque or clutch setting).

(70) The electrostatic clutch assembly **18** may include positive and negative brushes **371** (as shown in FIG. 5A). In one embodiment, the first electrode **304** and the second electrode **306** are operatively coupled to one or more positive and negative brushes **371**, respectively, through the shaft **302** and the clutch housing **310**. For example, the first electrode **304** is coupled to a negative brush **371** via the clutch housing **310** and the second electrode **306** is coupled to a positive brush **371** via the shaft **302**. In another embodiment, the first electrode **304** is coupled to a positive brush **371** via the clutch housing **310** and the second electrode **306** is coupled to a negative brush **371** via the shaft **302**. The positive and negative brushes **371** are stationary relative to the tool housing **12**. The positive and negative brushes **371** are configured to provide electrical current to charge the clutch housing **310**, the shaft **302** and/or their respective electrodes **304**, **306**.

(71) In one embodiment, one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is proximate to the front end of the housing **12** and is operatively coupled to the tool holder **22** (e.g., via the output spindle **68** of the power tool **10**) or to the output spindle **68** of the power tool **10**. In one embodiment, one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is configured to rotationally drive the end effector **22**, which is mounted to the output spindle **68** of the power tool **10** and is configured to receive the tool bit portion **23** therein.

(72) In one embodiment, one of the shaft **302** and the clutch housing **310** is integral with the output spindle **68** of the power tool **10**. In another embodiment, one of the shaft **302** and the clutch housing **310** is fixed to the output spindle **68** of the power tool **10**. In another embodiment, one of the shaft **302** and the clutch housing **310** is splined or otherwise removably coupled to the output spindle **68** of the power tool **10**. A person of ordinary skill in the art would readily understand that splined herein refers to securing/connecting one of the shaft **302** and the clutch housing **310** to the output spindle **68**, for example, by means of a spline arrangement.

(73) In one embodiment, the other of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is operatively coupled to the impact mechanism **402**, to the motor **15**, to the transmission **16**, or to the motor and transmission assembly **14**. In one embodiment, the other of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is configured to be driven by the impact mechanism **402**, the motor **15**, the transmission **16**, or the motor and transmission assembly **14**. The operatively coupled connection between the other of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** and the impact mechanism **402**, the motor **15**, the transmission **16**, or the motor and transmission assembly **14** are shown in and described in detail below with respect to FIGS. 2, 6-8, and 10-20.

(74) The clutch housing **310** may interchangeably referred to as can. The clutch housing **310** may be either the output or the input of the electrostatic clutch assembly **18**. The clutch housing **310** contains the first electrode **304**, the second electrode **306**, the dielectric layer **308** between the first electrode **304** and the second electrode **306**, and at least a portion of the shaft **302**. The clutch housing **310** may also contain an inner brush **365**, bearings **343**, **343L**, **343R**, insulation **355** that is

disposed around the bearings **343**, **343L**, **343R**, and an elastic member **335** that is configured to pretension the first electrode **304** and the second electrode **306** as will be described in detail below. (75) The clutch housing **310** may include a first portion **317** and an opposing second portion **319**. The clutch housing **310** may serve as either the input or the output of the electrostatic clutch assembly **18**. When the clutch housing **310** serves as the input of the electrostatic clutch assembly **18**, the shaft **302** serves as the output of the electrostatic clutch assembly **18**. When the clutch housing **310** serves as the output of the electrostatic clutch assembly **18**, the shaft **302** serves as the input of the electrostatic clutch assembly **18**.

(76) When the clutch housing **310** serves as the input of the electrostatic clutch assembly **18**, in the illustrated embodiment, the second portion **319** of the clutch housing **310** is operatively coupled to the motor transmission assembly **14**, to the motor **15**, to the transmission **16** or to the impact mechanism **402**, while the first portion **317** of the clutch housing **310** may be operatively connected to the one or more first electrodes **304** (i.e., each first electrode **304** of the plurality of electrode pairs **307**).

(77) When the clutch housing **310** serves as the output of the electrostatic clutch assembly **18**, in the illustrated embodiment, the second portion **319** of the clutch housing **310** is operatively coupled to the tool holder **22** (e.g., via the output spindle **68** of the power tool **10**) or to the output spindle **68** of the power tool **10**, while the first portion **317** of the clutch housing **310** may be operatively connected to the one or more first electrodes **304** (i.e., each first electrode **304** of the plurality of electrode pairs **307**).

(78) Referring to FIGS. 3-5A, the clutch housing **310** may include a front/first end portion **321**, a middle portion **323**, a rear/second end portion **325** and a shaft member **327**. The middle portion **323** may be configured to accommodate the plurality of electrode pairs **307** therein. The shaft member **327** may be pressed onto the front portion **321** of the clutch housing **310**. At least a portion of the front portion **321** may be configured to receive at least a portion of the shaft member **327**.

(79) The front portion **321** and the middle portion **323** may be integrally formed. The front portion **321** and the middle portion **323** may be separately formed and connected to each other (e.g., at their end portions). One end **329** of the middle portion **323** may be connected to the front portion **321**. An outer surface **351** of the front portion **321** may be configured to have connection portions **353** that are configured to engage/connect with connection portions **352** on the inner surface **333** of the middle portion **323**.

(80) The other end **331** of the middle portion **323** may be connected to the rear portion **325**. The middle portion **323** and the rear portion **325** may be threadably connected (e.g., via screwed connections **349**) to each other. The middle portion **323** and the rear portion **325** may be connected to each other by other releasably connection mechanisms as would be appreciated by a person of ordinary skill in the art.

(81) The front portion **321**, the middle portion **323**, and the rear portion **325** are constructed and arranged to accommodate the plurality of electrode pairs **307** therein. The middle portion **323** and the rear portion **325** are also constructed and arranged to receive at least portion of the shaft **302** therein.

(82) The front end portion **321**, the middle portion **323**, the rear end portion **325** and the shaft member **327** of the clutch housing **310** are all configured to provide a sealed arrangement for the electrostatic clutch assembly **18** and for the plurality of electrode pairs **307** accommodated therein.

(83) The inner surface **333** of the middle portion **323** may be configured to non-rotatably couple each of the plurality first electrodes **304** to the clutch housing **310**. The inner surface **333** of the middle portion **323** may also be configured to allow axial movement of each of the plurality first electrodes **304** relative to the clutch housing **310**. In one embodiment, each of the plurality first electrodes **304** are toothed on the inner surface **333** of the middle portion **323**.

(84) As shown in FIG. 3, the electrostatic clutch assembly **18** may also include an elastic member **335** (e.g., a silicone rubber member) that is disposed in the clutch housing **310** and is configured to

pretension the electrodes **304**, **306** when the electrodes **304**, **306** are received in the clutch housing **310**.

(85) The electrostatic clutch assembly **18** may also include two bearings **343** (e.g., right side bearing **343R** and left side bearing **343L**) that are positioned adjacent to the front portion **321** and the rear portion **325** of the clutch housing **310** when the electrostatic clutch assembly **18** is in its assembled configuration.

(86) The shaft **302** is configured to extend through at least portions of the front portion **321**, the middle portion **323**, and the rear portion **325** of the clutch housing **310** and the bearings **343**, **343R**, **343L** when the shaft **302** is received in the clutch housing **310**. Also, referring to FIG. 4, one end **337** of the shaft **302** is configured to be adjacent one end **339** of the shaft member **327** when the shaft **302** is received in the clutch housing **310**.

(87) Outer (circumferential) surface **341** of the shaft **302** may be configured to non-rotatably couple each of the plurality second electrodes **306** to the shaft **302**. The outer surface **341** of the shaft **302** is configured to allow axial movement of each of the plurality second electrodes **306** relative to the shaft **302**. In one embodiment, each of the plurality second electrodes **306** are toothed on the outer surface **341** of the shaft **302**.

(88) The shaft **302** may include a first portion **313** and an opposing second portion **315**. The shaft **302** may serve as either the input or the output of the electrostatic clutch assembly **18**. When the shaft **302** serves as the input of the electrostatic clutch assembly **18**, the clutch housing **310** serves as the output of the electrostatic clutch assembly **18**. When the shaft **302** serves as the output of the electrostatic clutch assembly **18**, the clutch housing **310** serves as the input of the electrostatic clutch assembly **18**.

(89) When the shaft **302** serves as the input of the electrostatic clutch assembly **18**, in the illustrated embodiment, the second portion **315** of the shaft **302** is operatively coupled to the motor transmission assembly **14**, to the motor **15**, to the transmission **16** or to the impact mechanism **402**, while the first portion **313** may be operatively connected to the one or more second electrodes **306** (i.e., each second electrode **306** of the plurality of electrode pairs **307**).

(90) When the shaft **302** serves as the output of the electrostatic clutch assembly **18**, the second portion **315** of the shaft **302** is operatively coupled to the tool holder **22** (e.g., via the output spindle **68** of the power tool **10**) or to the output spindle **68** of the power tool **10**, while the first portion **313** of the shaft **302** may be operatively connected to the one or more second electrodes **306** (i.e., each second electrode **306** of the plurality of electrode pairs **307**).

(91) Referring to FIGS. 5A and 5B, the electrostatic clutch assembly **18** may also include insulation portions **355** positioned around the bearings **343**. In one embodiment, the tool housing **12** is configured to insulate the electrostatic clutch assembly **18**. In one embodiment, sealing arrangements, including but not limited to, gaskets with O-rings may be used, additionally or alternatively, to seal the clutch housing **310** of the electrostatic clutch assembly **18**.

(92) The electrostatic clutch assembly **18** may also include a wired connection **359** for a shaft and brush connection **365** (i.e., an inner brush **365**), a disc **363** for the shaft and brush connection **365**, and a connection **369** from the disc **363** to the inner brush **365**. The shaft and brush connection **365** (i.e., the inner brush **365**) is configured to connect the shaft **302** and the corresponding brush **371**. The electrostatic clutch assembly **18** may also include a wired connection **361** for the clutch housing and brush connection and a disc **367** for the clutch housing and brush connection.

(93) Also, the electrostatic clutch assembly **18** may include other insulation portions **357** between the clutch housing **310** and the brush connection components **365**, **363**, **369**, and **367**. The electrostatic clutch assembly **18** may also include the brushes **371** and springs **373**.

(94) The brushes **371** are configured to provide electrical current/deliver power from the power source **102** (e.g., battery pack **26** of the power tool **10** via the wired connections **359**, **361**) to charge the clutch housing **310**, the shaft **302** and/or their respective electrodes **304**, **306**. The springs **373** are configured to maintain the connections between the wired connections **359**, **361** and their

respective brushes **371**. The discs **363**, **367** are configured to operatively connect the respective brushes **371** with either the clutch housing **310** or the shaft **302**. The inner brush **365** is configured to operatively connect the disc **363** to the shaft **302**. Thus, the power from the power source **102** (e.g., battery pack **26** of the power tool **10**) is delivered to charge the clutch housing **310**, the shaft **302** and/or their respective electrodes **304**, **306** via the wired connections **359**, **361**, the brushes **371** (using the springs **373** and the inner brush **365**), and the discs **363**, **367**.

(95) The control circuit or controller **50** of the power tool **10** may be disposed in the power tool housing **12** and may be operatively cooperable with the electrostatic clutch assembly **18**. The control circuit **50** may be interchangeably referred to as controller.

(96) The control circuit or controller **50** may be operatively cooperable with the motor and transmission assembly **14** and the electrostatic clutch assembly **18** to control electrical power/power delivery from the power source **102** to the motor and transmission assembly **14** and to the first and second electrodes **304**, **306**.

(97) The motor and transmission assembly **14** may have a second control circuit or controller that is independent of the first control circuit or controller **50** for the electrostatic clutch assembly **18** that is configured to control power delivery from the power source **102** to the motor and transmission assembly **14**. In one embodiment, the second control circuit or controller is operatively connected to the first control circuit or controller **50**. The second control circuit or controller is optional.

(98) In one embodiment, the control circuit or controller **50** may be implemented as a microcontroller. In other embodiments, the control circuit or controller **50** may be implemented as, be part of, or include an electronic circuit, an Application Specific Integrated Circuit (ASIC), a processor (shared, dedicated, or group) and/or memory (shared, dedicated, or group) that execute one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality. For ease of reference, the control circuit or controller **50** is referred to generally herein as a control circuit **50**; however, it should be understood that the term control circuit includes any of the aforementioned implementations.

(99) The control circuit **50** may be configured to cause a first voltage difference to be applied between the first electrode **304** and the second electrode **306**, generating a first attractive force between the first and second electrodes **304** and **306**, causing the shaft **302** and the clutch housing **310** to be coupled to each other when a torque on one of the shaft **302** and the clutch housing **310** is less than or equal to a first threshold value, and allowing one of the shaft **302** and the clutch housing **310** to rotationally slip relative to the other of the shaft **302** and the clutch housing **310** when the torque on one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** exceeds the first threshold value. That is, the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** are non-rotatably coupled to each other when the torque on one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is less than or equal to the first threshold value. For example, in one embodiment, the control circuit **50** may be configured to cause a first voltage to be applied to the first electrode **304** and a second voltage to be applied to the second electrode **306**, the first voltage is different from the second voltage.

(100) The control circuit **50** may be configured to cause a voltage difference or a voltage potential to be applied across the first and second electrodes **304**, **306**, generating a first attractive force between the first and second electrodes **304**, **306**. For example, a first voltage may be applied to the first electrode **304** and a second voltage with a polarity opposite a polarity the first voltage may be applied to the second electrode **306**, generating a first attractive force between the first and second electrodes **304** and **306**. That is, the control circuit **50** may cause the first voltage to be applied to the first electrode **304** and a different second voltage to be applied to the second electrode **306**. This causes the shaft **302** and the clutch housing **310** to rotate together when a torque on one of the shaft **302** and the clutch housing **310** is less than or equal to a first threshold value and allows one of the shaft **302** and the clutch housing **310** to rotationally slip relative to the other of the shaft **302** and the clutch housing **310** when the torque on one of the shaft **302** and the clutch housing **310** exceeds

the first threshold value, interrupting torque transmission between the shaft **302** and the clutch housing **310**.

(101) FIG. **3** shows operational connections between the control circuit **50**, the electrical power source **102** and the first and second electrodes **304**, **306** in accordance with an embodiment of the present patent application. FIGS. **5A** and **5B** show operational connections between the control circuit **50**, the electrical power source **102** and the first and second electrodes **304**, **306** in accordance with another embodiment of the present patent application (as described in detail throughout the present patent application). These operational connections may be configured to enable the control circuit **50**, disposed in the housing **12** and operatively cooperable with the electrostatic clutch assembly **18**, to control electrical power delivery from the electrical power source **102** to the first electrode **304** (e.g., via the clutch housing **310**) and to the second electrode **306** (e.g., via the shaft **302**), which in turn enables one of the first and second electrodes **304**, **306** to be a positively charged electrode and the other of the first and second electrodes **304**, **306** to be a negatively charged electrode. One of the first and second electrodes **304**, **306** may be positively charged when a first voltage is applied to one of the first and second electrodes **304**, **306** and the other of the first and second electrodes **304**, **306** may be negatively charged when a second voltage may be applied to the other of the first and second electrodes **304**, **306**. In one embodiment, the second voltage is different from the first voltage. In one embodiment, the second voltage has a polarity opposite a polarity the first voltage. In yet another embodiment, the second voltage is different from and has a polarity opposite to a polarity the first voltage. In one embodiment, each pair of the first and second electrodes **304**, **306** of the electrostatic clutch assembly **18** comprise a parallel-plate capacitor (i.e., two conducting plates **304**, **306**, with a dielectric therebetween). The plates are oppositely charged, as with any capacitor.

(102) The controller **50** may include a current sensing circuit and a position/rotation speed sensing circuit. The current sensing circuit may include a current sensor (e.g., a shunt resistor) that senses the amount of current being delivered to the motor **15** and provides a current sensing signal corresponding to the sensed current to the controller **50**. The position/rotation speed sensing circuit may include one or more position/rotation speed sensors that sense changes in the angular position of the motor output shaft and provides a signal corresponding to the angular rotation, speed, and/or acceleration of the motor **15** to the controller **50**. In one embodiment, the position sensors are Hall sensors that are already part of a brushless motor. For example, the power tool **10** may include a three-phase brushless motor, where the rotor includes a four-pole magnet, and there are three Hall sensors positioned at 120° intervals around the circumference of the rotor. As the rotor rotates, each Hall sensor senses when one of the poles of the four-pole magnet passes over the Hall sensor. Thus, the Hall sensors can sense each time the rotor, and thus the motor output shaft, rotates by an increment of 60°. In one embodiment, the rotation sensing circuit can use the signals from the Hall sensors to infer or calculate the amount of angular rotation, speed, and/or acceleration of the rotor. For example, the rotation sensing circuit includes a clock or counter that counts the amount of time or the number of counts between each 60° rotation of the rotor. The controller **50** can use this information to calculate or infer the amount of angular rotation, speed, and/or acceleration of the motor **15**. In one embodiment, the current sensing circuit, the current sensor, the position/rotation sensing circuit, and the position/rotation sensors are optional.

(103) The power tool/system **10** may include a sensor configured to sense whether the electrostatic clutch assembly **18** is in a fully engaged mode, a fully disengaged mode, or is slipping and output a signal to the controller **50**. In one embodiment, the controller **50**, in response to the received signal from the sensor, is configured to stop the rotation of the motor and transmission assembly **14**. The sensor may comprise one or more of a current sensor, a position sensor, and a rotational motion sensor. The power tool system **10** may include a sensor that senses when the electrostatic clutch assembly **18** slips or rotates (e.g., a current sensor or a rotational motion sensor as described above) and that causes the control circuit/controller **50** to de-energize the electrostatic films/layers after the

electrostatic clutch assembly **18** slips or rotates.

(104) Referring to FIGS. **2** and **6-9B**, in an embodiment, the power tool **10** comprises a drill or drill/driver that includes a housing **12**, a motor **15**, a planetary transmission **16**, an end effector **22**, an electrostatic clutch assembly **18**, and a control circuit **50**. One of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is configured to be rotationally driven by the motor **15**. One of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is operatively connected to the motor **15** via an input shaft/member that is driven by the motor **15**. The motor assembly **14** may interchangeably referred to as the motor and transmission assembly and includes the motor **15** and the planetary transmission **16**.

(105) One of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** may be non-rotatably coupled to an output carrier **627** of the planetary transmission **16**. That is, one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** and the output carrier **627** of the planetary transmission **16** rotate together. In one embodiment, one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is fixed to the output carrier **627** of the planetary transmission **16**. In another embodiment, one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is integral with the output carrier **627** of the planetary transmission **16**. In yet another embodiment, one of the shaft **302** and the clutch housing **310** is splined to the output carrier **627** of the planetary transmission **16** by using a spline arrangement.

(106) For example, a person of ordinary skill in the art would appreciate the planetary transmission **16** may include a multi-stage planetary gear set. Each stage includes an input sun gear, a plurality of planet gears meshed with the sun gear and pinned to a rotatable planet carrier, and a ring gear meshed with and surrounding the planet gears. For each stage of the multi-stage planetary gear set, if a ring gear is rotationally fixed relative to the housing **12**, the planet gears orbit the sun gear when the sun gear rotates, transferring power at a reduced speed to their planet carrier, thus causing a speed reduction through that particular stage. If a ring gear is allowed to rotate relative to the housing **12**, then the sun gear causes the planet carrier to rotate at the same speed as the sun gear, causing no speed reduction through that stage. By varying which one or ones of the stages have the ring gears are fixed against rotation, the total amount of speed reduction through the transmission **16** can be controlled, and thus the speed setting of the transmission **16** (e.g., among high, medium, and low) can be adjusted.

(107) For example, referring to FIG. **2**, the planetary transmission **16** may include a three-stage planetary gear set. The first stage planetary gear set includes an input sun gear **607**, a plurality of planet gears **609** meshed with the sun gear **607** and pinned to a rotatable planet carrier **611**, and a ring gear **613** meshed with and surrounding the planet gears **609**. The input sun gear **607** is fixed to the output motor shaft **603** of the motor **15**. The ring gear **613** is rotationally fixed relative to the housing **12** such that the planet gears **609** orbit the sun gear **607** when the sun gear **607** rotates, transferring power at a reduced speed to their planet carrier **611**, thus causing a speed reduction through the first stage planetary gear set.

(108) Similarly, referring to FIG. **2**, the second stage planetary gear set includes a sun gear **615**, a plurality of planet gears **617** meshed with the sun gear **615** and pinned to a rotatable planet carrier **619**, and a ring gear **621** meshed with and surrounding the planet gears **617**. The sun gear **615** is operatively connected to the first stage planet carrier **611**. The ring gear **621** is rotationally fixed relative to the housing **12** such that the planet gears **617** orbit the sun gear **615** when the sun gear **615** rotates, transferring power at a reduced speed to their planet carrier **619**, thus causing a speed reduction through the second stage planetary gear set.

(109) Further similarly, referring to FIG. **2**, the third stage planetary gear set includes a sun gear **623**, a plurality of planet gears **625** meshed with the sun gear **623** and pinned to the rotatable planet carrier **627**, and a ring gear **629** meshed with and surrounding the planet gears **625**. The sun gear **623** is operatively connected to the second stage planet carrier **619**. The ring gear **629** is

rotationally fixed relative to the housing **12** such that the planet gears **625** orbit the sun gear **623** when the sun gear **623** rotates, transferring power at a reduced speed to their planet carrier **627**, thus causing a speed reduction through the through stage planetary gear set. The third second stage planet carrier **627** may be interchangeably referred to as output carrier of the three-stage planetary gear set.

(110) Although three stage planetary gear set is described here, the number of stages in the planetary gear set may vary and may depend on the application. In one embodiment, the planetary gear set may include one stage planetary gear set. In another embodiment, the planetary gear set may include two stage planetary gear set.

(111) The planetary transmission **16** may be configured to be rotationally driven by the motor **15**. The planetary transmission **16** includes the output planet carrier **627** as a rotational output of the transmission **16**. As shown in FIG. 2, the electrostatic clutch assembly **18** is disposed in the power tool housing **12** between the planetary transmission **16** and the end effector **22**. The electrostatic clutch assembly **18** includes the input member (e.g., one of the shaft **302** and the clutch housing **310**) that is non-rotatably coupled to the first electrode **304**, the output member (e.g., other of the shaft **302** and the clutch housing **310**) that is non-rotatably coupled to the second electrode **306**, and the dielectric layer **308** arranged between the first and second electrodes **304**, **306**. The control circuit **50** is disposed in the power tool housing **12** and is operatively cooperable with the electrostatic clutch assembly **18** to control power delivery from the power source **102** to the first electrode **304** and the second electrode **306**.

(112) The input member (e.g., one of the shaft **302** and the clutch housing **310**) of the electrostatic clutch assembly **18** is operatively coupled to the output planet carrier **627** of the transmission **16** to rotate with the output planet carrier **627** and the output member (e.g., one of the shaft **302** and the clutch housing **310**) is operatively coupled to the end effector **22** to rotate with the end effector **22**. The control circuit **50** causes a first voltage to be applied to the first electrode **304** and a second voltage to be applied to the second electrode **306**, the first voltage being different from the second voltage, generating a first attractive force between the first and second electrodes **304**, **306**, causing the input member (e.g., one of the shaft **302** and the clutch housing **310**) of the electrostatic clutch assembly **18** and the output member (e.g., other of the shaft **302** or the clutch housing **310**) of the electrostatic clutch assembly **18** to be coupled to each other when a torque on the output member (e.g., other of the shaft **302** and the clutch housing **310**) is less than or equal to a first threshold value, and allowing the output member (e.g., other of the shaft **302** and the clutch housing **310**) of the electrostatic clutch assembly **18** to rotationally slip relative to the input member (e.g., one of the shaft **302** and the clutch housing **310**) of the electrostatic clutch assembly **18** when the torque on the output member (e.g., other of the shaft **302** and the clutch housing **310**) of the electrostatic clutch assembly **18** exceeds the first threshold value.

(113) As shown in FIGS. 6-8, the input member (e.g., one of the shaft **302** and the clutch housing **310**) of the electrostatic clutch assembly **18** may be fixed to the output carrier **627** of the planetary transmission **16**. Also, as shown in FIGS. 6-8, the input member (e.g., one of the shaft **302** and the clutch housing **310**) of the electrostatic clutch assembly **18** may be integral with the output carrier **627** of the planetary transmission **16**. As shown in FIGS. 10-11, the input member (e.g., one of the shaft **302** and the clutch housing **310**) of the electrostatic clutch assembly **18** may be splined to the output carrier **627** of the planetary transmission **16**, for example, using a splined arrangement.

(114) In one embodiment, the shaft **302** of the electrostatic clutch assembly **18** is fixed to/integral with or is splined to the output carrier **627** of the planetary transmission **16**, while the clutch housing **310** of the electrostatic clutch assembly **18** is fixed to/integral with or is splined to the output spindle **68** of the power tool **10**.

(115) In another embodiment, as illustrated in FIGS. 6-8, the clutch housing **310** of the electrostatic clutch assembly **18** is fixed to/integral with or is splined to the output carrier **627** of the planetary transmission **16**, while the shaft **302** of the electrostatic clutch assembly **18** is fixed to/integral with

or is splined to the output spindle **68** of the power tool **10**.

(116) As shown in FIGS. **6-8**, when the clutch housing **310** of the electrostatic clutch assembly **18** is configured to serve as the output carrier, the rear portion **325** of the electrostatic clutch assembly **18** includes openings **637** that are configured to receive pins **639**, which are configured to pin the plurality of planet gears **625** of the third stage planetary gear set to the output carrier/clutch housing **310** of the electrostatic clutch assembly **18**. FIGS. **9A-9B** show the output carrier/clutch housing **310** of the electrostatic clutch assembly **18** with the plurality of planet gears **625** of the third stage planetary gear set being pinned thereto.

(117) Referring to FIGS. **10-11**, when the clutch housing **310** of the electrostatic clutch assembly **18** is splined to the transmission **16**, the rear portion **325** of the electrostatic clutch assembly **18** includes a splined connection **643** that is configured to be received in an (correspondingly shaped) opening **645** of the output carrier **627** of the transmission **16** so as to operatively connect the electrostatic clutch assembly **18** to the transmission **16**.

(118) The slip torque for the electrostatic clutch assembly may be more accurate and reproducible in this embodiment, in which one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is mounted to the output carrier **627** of the planetary transmission **16** and the other of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is operatively connected to the output spindle **68** of the power tool **10**. In this embodiment, the electrostatic clutch assembly **18** is close to the output spindle **68** of the power tool **10** with no intervening transmission components to introduce slack or backlash.

(119) Also, referring to FIGS. **6-8**, the power tool **10** may also include insulating housing **631** that is configured to insulate the electrostatic clutch assembly **18**. The insulating housing **631** may be part of the housing **12** of the power tool **10**. The insulating housing **631** may be disposed adjacent to and may surround the electrostatic clutch assembly **18** so as to insulate the electrostatic clutch assembly **18**.

(120) Referring to FIGS. **6-8**, the power tool **10** may also include an insulating front housing **633**, a bearing support ring/front portion **641** and an output bearing **635** that are configured to insulate the shaft **302**/output member of the electrostatic clutch assembly **18** near a front portion of the electrostatic clutch assembly **18**. When the electrostatic clutch assembly **18** is assembled, the insulating front housing **633**, the front portion **641**, the output bearing **635** and the insulating housing **631** together completely insulate the electrostatic clutch assembly **18** on the front and the sides of the electrostatic clutch assembly **18**.

(121) The insulating front housing **633**, the front portion **641**, the output bearing **635** may also be configured to operatively connect the output (e.g., the shaft **302**) of the electrostatic clutch assembly **18** to the output spindle **68**. In one embodiment, the front portion **641** of the power tool may be referred to as a carrier or bushing.

(122) In another embodiment of a drill or drill/driver **1200**, as shown in FIG. **12**, the electrostatic clutch assembly **1218** may be disposed between the motor **1215** and the transmission **1216**. That is, the electrostatic clutch assembly **1218** is disposed between the output shaft/member **1203** of the motor **1215** and an input member (e.g., a sun gear or a first stage sun gear) **1247** of the planetary transmission **1216**. The drill/driver **1200**, the electrostatic clutch assembly **1218**, the motor **1215**, the transmission **1216**, the output shaft/member **1203**, the input member (e.g., a sun gear or a first stage sun gear) **1247** of the planetary transmission **1216**, the end effector **1222**, the output spindle **1268**, the shaft **1202**, the clutch housing **1210**, and the motor and transmission assembly **1214** are substantially similar to the drill/driver **10**, the electrostatic clutch assembly **18**, the motor **15**, the transmission **16**, the output shaft/member **603**, the input member (e.g., a sun gear or a first stage sun gear) **647** of the planetary transmission **16**, the end effector **22**, the output spindle **68**, the shaft **302**, the clutch housing **310**, and the motor and transmission assembly **14** described in detail in the present patent application.

(123) In one embodiment, one of the shaft **1202** and the clutch housing **1210** is fixed to a first stage

sun gear **1247** of the planetary transmission **1216** and the other of the shaft **1202** and the clutch housing **1210** is splined to the output shaft **1203** of the motor **1215**. In another embodiment, one of the shaft **1202** and the clutch housing **1210** is fixed to the first stage sun gear **1247** of the planetary transmission **1216** and the other of the shaft **1202** and the clutch housing **1210** is fixed to or integral with the output shaft **1203** of the motor **1215**.

(124) The slip torque accuracy in this embodiment, in which the electrostatic clutch assembly **1218** is disposed between the motor **1215** and the transmission **1216**, may not be as accurate as in the above-described embodiment, in which the electrostatic clutch assembly **1218** is disposed between the motor and transmission assembly **1214** (including the motor **1215** and/or the transmission **1216**) and the end effector **1222**. In this embodiment, the electrostatic clutch assembly **1218** is located before the transmission **1216** (e.g., in a direction starting with the motor **1215** and moving towards the output spindle **1268**) and the transmission **1216** can introduce slack or backlash. However, this embodiment also may require less holding torque because the torque before the transmission **1216** is lower than the torque at the output of the transmission **1216**.

(125) Referring to FIGS. **13-14**, in another embodiment of a drill or drill/driver **10**, the electrostatic clutch assembly **18** may be configured to selectively couple and decouple the output ring gear of a planetary transmission to the tool housing. In this embodiment, one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** is non-rotatably coupled to an output stage ring gear **649** of the planetary transmission **16**. The other of the shaft **302** and the clutch housing **310** is non-rotatably coupled to the tool housing. The gears in the planetary transmission **16** are configured to be rotationally driven by the motor **15**.

(126) The electrostatic clutch assembly **18** includes a first member (e.g., one of the shaft **302** and the clutch housing **310**) that is non-rotatably coupled to the first electrode **304** and a second member (e.g., the other of the shaft **302** and the clutch housing **310**) that is non-rotatably coupled to a second electrode **306**, and that dielectric layer **308** that is arranged between the first and second electrodes **304**, **306**. The control circuit **50** is disposed in the power tool housing **12** and operatively cooperable with the electrostatic clutch assembly **18** to control power delivery from the power source **102** to the first electrode **304** and the second electrode **306**.

(127) The control circuit **50** causes a first voltage to be applied to the first electrode **304** and a second voltage to be applied to the second electrode **306**, the first voltage being different from the second voltage, generating a first attractive force between the first and second electrodes **304**, **306**, causing the ring gear **649** to remain stationary relative to the power tool housing **12**, enabling torque transmission from the planetary transmission **16** to the end effector **22**. When a torque on the end effector exceeds a threshold value that corresponds to the holding force between the first and second electrodes, the electrodes slip relative to one another, which allows the ring gear **649** to rotate relative to the power tool housing **12**, interrupting torque transmission from the planetary transmission **16** to the end effector **22**.

(128) FIGS. **13-14** illustrate how the electrostatic clutch assembly **18** acts on the ring gear **649** of the output stage of the planetary transmission **16**. One of the first and second electrodes **304**, **306** is coupled to the clutch housing **310**, which is fixed to the housing **12**. In the ring gear clutch embodiment, one of the electrodes never rotates. The other of the first and second electrodes **304**, **306** is coupled to a member **651** that is coupled to the output stage ring gear **649**.

(129) When the output torque on the output spindle **68** is below the clutch setting, the first and second electrodes **304**, **306** do not rotate relative to one another, and the ring gear **649** is held stationary relative to the housing **12**, which allows torque to be transmitted through the output stage of the planetary transmission **16**. When the torque on the output spindle **68** exceeds the set value, the first and second electrodes **304**, **306** slip and the member **651** rotates, which allows the ring gear **649** to rotate, interrupting torque transmission through the output stage of the planetary transmission **16**. That is, the ring gear **649** is configured to rotate when the torque on the output spindle **68** exceeds the set value.

(130) This embodiment of the electrostatic clutch assembly **18** requires only one set of electrodes **304, 306** to be energized by brushes because only one set of electrodes **304, 306** is movable relative to the tool housing. However, this design of the electrostatic clutch assembly **18** may be less accurate than the embodiment of the electrostatic clutch assembly **18** in FIGS. 2 and 6-11, in which one of the shaft **302** and the clutch housing **310** is mounted on the output carrier **627** of the planetary transmission **16** because it is acting on the output ring gear **649** of the planetary transmission **16** and not the output carrier **627** of the planetary transmission **16**, so there may be some backlash or slack introduced.

(131) Referring to FIG. 14, the power tool **10** may also include a seal member (e.g., grease seal) **653** and a compression member **655** on one side of the plurality of electrode pairs **307** of the electrostatic clutch assembly **18** and may include bearing **343** and a stop plate **657** on the other side of the plurality of electrode pairs **307** of the electrostatic clutch assembly **18**. Also, FIG. 14 shows another configuration in which the brush **371** is connected to the clutch housing **310**.

(132) In other embodiments, referring to FIGS. 15-20, an electrostatic clutch assembly **418, 518, 818** may be configured for an impact tool **410, 510, 810**, in which the electrostatic clutch assembly **418, 518, 818** is configured to limit a torque transmitted to the output spindle **468, 568, 868** of the impact tool **410, 510, 810** to a maximum threshold value.

(133) In these embodiments, an impact power tool **410, 510, 810** may generally include a power tool housing **412, 512, 812**, a motor **415, 515, 815** that is received in the power tool housing **412, 512, 812** and configured to be powered by a power source **4102, 5102, 8102** and an output spindle **468, 568, 868** that is at least partially received in and rotatable relative to the power tool housing **412, 512, 812**, an end effector **422, 522, 822** coupled to the output spindle **468, 568, 868** an electrostatic clutch assembly **418, 518, 818** and one or more control circuits/controllers **450, 550, 850** configured to control power delivery and operation of the motor **415, 515, 815** and the electrostatic clutch assembly **418, 518, 818**. The electrostatic clutch assembly **418, 518, 818** includes a shaft **4302, 5302, 8302** a first electrode **4304, 5304, 8304** and a second electrode **4306, 5306, 8306** axially arranged along an axial direction of the shaft **4302, 5302, 8302**. In one embodiment, the end effector **422, 522, 822** of the impact power tool **410, 510, 810** may be the same as the end effector **22** in the power drill **10**. In another embodiment, the end effector **422, 522, 822** of the impact power tool **410, 510, 810** may be different than the end effector **22** in the power drill **10**.

(134) The first electrode **4304, 5304, 8304** and the second electrode **4306, 5306, 8306** are substantially parallel to each other. The electrostatic clutch assembly **418, 518, 818** includes a clutch housing **4310, 5310, 8310** containing the first electrode **4304, 5304, 8304**, the second electrode **4306, 5306, 8306**, and at least a portion of the shaft **4302, 5302, 8302**. The first electrode **4304, 5304, 8304** is non-rotatably coupled to the clutch housing **4310, 5310, 8310**. The second electrode **4306, 5306, 8306** is non-rotatably coupled to the shaft **4302, 5302, 8302**. At least one of the electrodes is axially moveable relative to the other electrode, as described above. In one embodiment, the power tool housing, the motor, the control circuit/controller, the electrostatic clutch assembly, the output spindle and the power source in the impact power tools may be similar to the power tool housing **12**, the motor **15**, the controller/control circuit **50**, the transmission **16**, the electrostatic clutch assembly **18**, the output spindle **68**, and the power source **102**, respectively as shown and described in other embodiments of the present patent application, and thus these will not be shown and described in detail here.

(135) The transmission **416, 516, 816** in the impact tool **410, 510, 810** may not necessarily be similar to the transmission in a drill/driver. For example, a drill/driver may have a multi-stage planetary transmission, while an impact tool **410, 510, 810** usually has only a single stage transmission. Also, the planet carrier in the impact tool **410, 510, 810** may generally be integral with the cam shaft (even though that is also part of the impact mechanism). The impact tool **410, 510, 810** may be referred to as power impact tool and the drill/driver **10** may be referred to as

power drill/driver.

(136) The impact power tool **410, 510, 810** also includes impact mechanism **402, 502, or 802** that is operatively coupled with the motor **415, 515, 815** and configured to be driven thereby. The impact mechanism **402, 502, or 802** is configured to generate/apply intermittent rotational impacts that are transmitted to the output spindle **468, 568, 868** of the power tool **410, 510, 810**. The impact mechanism **402, 502, or 802** may be a rotational impact mechanism. The electrostatic clutch assembly **418, 518, 818** is disposed in the power tool housing **412, 512, 812** and between the impact mechanism **402, 502, 802** and the output spindle **468, 568, 868**. The control circuit **450, 550, 850** is disposed in the power tool housing **412, 512, 812** and operatively cooperable with the electrostatic clutch assembly **418, 518, 818**. The control circuit **450, 550, 850** causes a first voltage to be applied to the first electrode **4304, 5304, 8304** and a second voltage to be applied to the second electrode **4306, 5306, 8306**, the first voltage is different from the second voltage, generating a first attractive force between each of the first and second electrodes **4304, 5304, 8304, 4306, 5306, 8306**, causing the shaft **4302, 5302, 8302** to rotate together with the clutch housing **4310, 5310, 8310** when a torque on the shaft **4302, 5302, 8302** is less than or equal to a first threshold value and causing the shaft **4302, 5302, 8302** to rotationally slip relative to the clutch housing **4310, 5310, 8310** when the torque on the shaft **4302, 5302, 8302** exceeds the first threshold value.

(137) As shown in FIGS. **15-17**, in an embodiment, a power tool **410** includes an impact mechanism **402** that comprises a Pott-type impact mechanism **402P**. One of the shaft **4302** or the clutch housing **4310** of the electrostatic clutch assembly **418** is operatively coupled to the output spindle **468**. The other of the shaft **4302** and the clutch housing **4310** is operatively coupled to an output of the Pott-type impact mechanism **402P**. The Pott-type impact mechanism **402P** comprises a cam shaft **437** rotationally driven by the output of the transmission (e.g., by the output carrier of a planetary transmission), a hammer **431** received at least partially over the cam shaft **437** and operatively coupled to the cam shaft **437** to be movable axially and rotationally relative to the cam shaft **437**, an anvil **433**, and a spring **435** axially biasing the hammer **431** toward the anvil **433**. When the torque encountered by the output spindle **468** is at or below a first threshold (which corresponds to the mechanical interaction between the components of the impact mechanism **402P**), the spring **435** maintains the hammer **431** in engagement with the anvil **433**, so that the cam shaft **437**, the hammer **431**, and the anvil **433** rotate in unison. When the torque encountered by the output spindle **468** exceeds the first threshold, the hammer **431** moves axially and rotationally along the cam shaft **437** to apply rotational impacts to the anvil **433**. The torque applied to the anvil when applying rotational impacts exceeds the torque applied when the hammer and the anvil rotate in unison and exceeds, and often greatly exceeds, the first threshold at which the impact tool begins impacting. For example, U.S. Pat. No. 10,406,662, which is incorporated by reference in the present patent application in its entirety, discloses an exemplary Pott-type impact mechanism.

(138) In the illustrated embodiment, the electrostatic clutch assembly **418** is disposed in the power tool housing **412** between the anvil **433** and the output spindle **468**, with the clutch housing **4310** fixedly coupled to the anvil **433** and the shaft **4302** fixedly coupled to the output spindle **468**. In an alternative embodiment, the shaft **4302** may be fixedly coupled to the anvil **433** and the clutch housing **4310** may be fixedly coupled to the output spindle **468**. As described further below, the electrostatic clutch assembly **418** will slip when the torque applied to the anvil **433** by the hammer **431** is above a second torque threshold (which corresponds to the holding force between the electrodes **4304, 4306**, as described above). This will limit the amount of torque transmitted from the anvil **433** to the output spindle **468** to be less than or equal to the second torque threshold.

(139) Referring to FIGS. **16-17**, the power impact tool **410** may also include an insulating front housing **439** and an output bearing **441** that are configured to insulate the electrostatic clutch assembly **418** when the electrostatic clutch assembly **418** is assembled. The power impact tool **410** may also include a nose cone **443** disposed between the electrostatic clutch assembly **418** and the anvil **433** as shown in FIG. **16**. The nose cone **443** may be configured to enable connection

between the electrostatic clutch assembly **418** and (e.g., the anvil **433**) of the Pott-type impact mechanism **402P**.

(140) As shown in FIGS. **18-19**, in another embodiment, the power tool **510** includes an impact mechanism **502** that comprises a twin hammer impact mechanism **502TH**. One of the shaft **5302** and the clutch housing **5310** is operatively coupled to an output of the twin hammer impact mechanism **502TH**. The other of the shaft **5302** or the clutch housing **5310** of the electrostatic clutch assembly **518** is operatively coupled to the output spindle **568**. In one embodiment, the motor **515** is a pneumatic motor. In another embodiment, the motor **515** is an electric motor. The construction and operation of the twin hammer impact mechanism is well-known to a person of ordinary skill in the art. For example, U.S. Pat. No. 6,491,111, which is incorporated by reference in the present patent application in its entirety provides an exemplary twin hammer impact mechanism. As described in U.S. Pat. No. 6,491,111, the twin hammer impact mechanism **502TH** includes a hollow cage or carrier member **520** rotatably drivable by the motor, an anvil **524** received in the carrier member **520** to allow rotation of the hollow cage or carrier member **520** relative to anvil **524**, and hollow hammer members **540** mounted around the anvil **524**.

(141) In the illustrated embodiment, the electrostatic clutch assembly **518** is disposed in the power tool housing **512** between the anvil **524** and the output spindle **568**, with the clutch housing **5310** fixedly coupled to the anvil **524** and the shaft **5302** fixedly coupled to the output spindle **568**. In an alternative embodiment, the shaft **5302** may be fixedly coupled to the anvil **524** and the clutch housing **5310** may be fixedly coupled to the output spindle **568**. As described further below, the electrostatic clutch assembly **518** will slip when the torque applied to the anvil **524** by the hammer **540** is above a second torque threshold (which corresponds to the holding force between the electrodes, as described above). This will limit the amount of torque transmitted from the anvil **524** to the output spindle **568** to be less than or equal to the second torque threshold.

(142) As shown in FIG. **20**, in another embodiment, the power impact tool **810** includes an impact mechanism **802** that comprises a hydraulic or oil impulse impact mechanism **802HI**. One of the shaft **8302** and the clutch housing **8310** of the electrostatic clutch assembly **818** is operatively coupled to an output shaft **820** of the hydraulic/oil impulse impact mechanism **802HI**. The construction and operation of the hydraulic/an oil impulse impact mechanism is well-known to a person of ordinary skill in the art. For example, U.S. Pat. No. 11,260,515, which is incorporated by reference in the present patent application in its entirety provides an exemplary oil pulse impact mechanism. As described in U.S. Pat. No. 11,260,515, the impact mechanism **802HI** includes a generally cylindrical hammer **880** containing fluid such as oil and coupled for rotation with an output carrier **819** of a planetary transmission **816** that is driven by a motor **823**. In other embodiments, the hammer **880** may be driven directly by the motor or by another component of a planetary or other type of transmission. The hammer **880** also includes an inwardly protruding impact member or lug (not shown). Received in the hammer **880** is a cam shaft **883** and an anvil shaft **881** that is coupled for rotation with the output shaft **820** of the oil pulse impact mechanism. The anvil shaft **881** includes a pair of impact members or blades (not shown) configured to be radially moveable relative to the anvil, and to be struck by the hammer lugs when in their radially outward position. Upon activation of the motor, torque from the electric motor is transferred to the hammer **880** via the transmission **816**, causing the hammer **880** and the cam shaft **883** to rotate in unison relative to the anvil shaft **881** until the impact members on the hammer **880** impact the respective blades on the anvil shaft **881** to deliver a rotational impact to the anvil shaft **881**. Further details regarding the construction and operation of the impact mechanism **802HI** can be found in the aforementioned U.S. Pat. No. 11,260,515.

(143) Referring to FIGS. **25A-25B**, in one implementation of the above described embodiments of impact tools with an electrostatic clutch assembly, the electrostatic clutch assembly **418**, **518**, **818** is configured to receive an input torque from one of the above-described impact mechanisms **402**, **502**, **802** (which may vary over time based on joint characteristics, power delivered to the motor,

and characteristics of the impact mechanism) and limit an amount of the input torque that is delivered to the output spindle to be less than the second torque threshold (i.e., the output torque delivered to the output spindle is limited to be no more than the second torque threshold). The input torque (i.e., measured in Nm) or the output torque (i.e., measured in Nm) is shown on left-hand side Y-axis of the graphs, while time (i.e., measured in seconds) is shown in the X-axis of the graphs. FIG. 25A illustrates the input torque delivered from the hammer to the anvil by rotary impacts of the hammer, with each impulse labeled as A1, A2, etc. FIG. 25B illustrates the output torque transmitted from the anvil to the output spindle with each impulse labeled as B1, B2, etc. The electrostatic clutch assembly limits the amount of torque transmitted to be less than or equal to the second threshold torque (indicated by line C), by slipping when the torque exceeds the second threshold C, effectively clipping the maximum torque delivered to the spindle. This second torque threshold may be preset in the power tool or may be user adjustable. Further, as shown by impulse A8 in FIG. 25A and bar B8 in FIG. 25B, if the input torque delivered to the anvil is less than the second torque threshold C, then the torque delivered from the anvil to the output spindle will be the same as the input torque, since the electrostatic clutch will not slip. As shown in FIGS. 25A and 25B, in this embodiment, the second threshold is constant and/or fixed over time.

(144) Referring to FIGS. 26A-26B, in another implementation of in one implementation of the above described embodiments of impact tools with an electrostatic clutch assembly, the second torque threshold is configured to be adjusted dynamically and/or vary over time. The second threshold value may be a function of time $f(t)$ as shown in FIG. 26B. For example, as shown in FIG. 26B, the second threshold value may be constant until time t_1 and then may decrease logarithmically after time t_1 . The function $f(t)$ is not limited to the illustrated function and may comprise any function or method of dynamically varying the second torque threshold over time. This function $f(t)$ may be preset in the power tool or may be user adjustable. Further, as shown by impulse B8 in FIG. 26A and impulse B8 in FIG. 25B, if the input torque delivered to the anvil is less than the torque threshold at a particular time t as indicated by the function $f(t)$, then the torque delivered from the anvil to the output spindle will be the same as the input torque, since the electrostatic clutch will not slip.

(145) FIG. 21 shows a system architecture for one of more of the above-described power tools, which collectively will be referred to as a power tool 710 for purposes of this discussion. In one embodiment, referring to FIG. 21, the power tool 710 includes a power source 702, a motor 715 (of the motor and transmission assembly 714, the motor and transmission assembly 714 may also include transmission 716) that is configured to be driven by the power source 702, an end effector 722 that is configured to perform an operation on a workpiece (not shown), and an electrostatic clutch assembly 718 that is arranged between the motor 715 and the end effector 722. The electrostatic clutch assembly 718 is similar to the electrostatic clutch 18, and includes a first electrode 704 and a second electrode 706 that are substantially parallel and adjacent to each other, and a dielectric layer 708 arranged between the first electrode 704 and the second electrode 706.

(146) In one embodiment, the power source 702 may include the (removable) power tool battery 726. The power tool 710 may also include a high voltage power supply 2106. The high voltage power supply 2106 may have a range of voltage between 0 volts and 2 Kilovolts.

(147) The motor and transmission assembly 714 include the motor 715 and the transmission 716. The electrostatic clutch assembly 718 is configured to couple the output of the motor and transmission assembly 714 to the output spindle 768 and the end effector 722 of the power tool 710. The chuck/bit holder 722 is configured to provide the output shaft/spindle 768 fixturing to couple bits/tools/accessories, to be applied to work surface/fasteners.

(148) The power tool 710 may include a controller or control circuit 750 that is operatively cooperable with the electrostatic clutch assembly 718 to control power delivery from the power source 702 to the first electrode 704 and the second electrode 706. The control circuit 750 causes a first voltage to be applied to the first electrode 704 and a second voltage to be applied to the second

electrode **706**, the first voltage being different from the second voltage generating a first attractive force between the first electrode **704** and the second electrode **706**. The control circuit **750** may include motor control circuit that is configured to provide intelligent control of the motor speed of the motor **715** and intelligent control of the high voltage power supply.

(149) The power tool **710** also includes a switch circuit **2102** that is operatively cooperable with the control circuit **750**. The switch circuit **2102** is configured to selectively apply power to the first electrode **704** and the second electrode **706**. In one embodiment, the switching circuit **2102** may be a H-Bridge circuit. The H-Bridge circuit includes semiconductor switches/switching elements **2108** (**2108A-D**) that are used to selectively apply power to the first electrode **704** and the second electrode **706**. A person of ordinary skill in the art would readily appreciate that the H-Bridge is an electronic circuit that is configured to switch the polarity of a voltage applied to a load and that the H-Bridge generally includes four switches **2108** (**2108A-D**) that are configured as the branches of a letter “H” and the load connected as the cross-bar.

(150) The switch circuit **2102** may be configured to change/switch a voltage polarity between a) the first electrode **704** being at the first voltage at a first polarity and the second electrode **706** being at the second voltage at a second polarity, and b) the first electrode **704** being at the first voltage at the second polarity and the second electrode **706** being at the second voltage at the first polarity. The switch circuit **2102** may further be configured to drain residual voltages from the first and second electrodes **704**, **706**.

(151) For example, in one embodiment, when the switches **2108A** and **2108C** are activated/closed, while the switches **2108B** and **2108D** are opened, a first voltage with a first polarity (e.g., one of the positive polarity and the negative polarity) is applied to the first electrode **704** and a second voltage with a second polarity (e.g., the other of the positive polarity and the negative polarity) is applied to the second electrode **706**. When switches **2108B** and **2108D** are activated/close, while the switches **2108A** and **2108C** are opened, the first voltage with the second polarity (e.g., the other of the positive polarity and the negative polarity) is applied to the first electrode **704** and the second voltage with the first polarity (e.g., one of the positive polarity and the negative polarity) is applied to the second electrode **706**.

(152) When the switches **2108C** and **2108D** are closed/activated, while the switches **2108A** and **2108B** are opened, the H-bridge/switch circuit **2102** may be configured to provide additional operation mode in which the electrostatic clutch assembly **718** brakes and operates (e.g., free run) until a frictional stop. When the switches **2108A** and **2108B** are closed/activated, while the switches **2108C** and **2108D** are opened, the H-bridge/switch circuit **2102** may be configured to provide additional operation mode in which the electrostatic clutch assembly **718** brakes and operates (e.g., free run) until a frictional stop.

(153) When the switches **2108A** and **2108D** are closed/activated at the same time, this would cause a short circuit on an input voltage source. Similarly, when the switches **2108B** and **2108C** are closed/activated at the same time, this would cause a short circuit on the input voltage source.

(154) The power tool **710** may further comprise a sense circuit **2104** that is configured to sense a characteristic of the electrostatic clutch assembly **718** and provide the sensed characteristic as an input to the control circuit **750**. The control circuit **750** is configured to determine, based on the input, the first voltage to be applied to the first electrode **704** and the second voltage to be applied to the second electrode **706**. The control circuit **750** is configured to transmit a signal to the switch circuit **2102**, the signal including the determined first voltage and/or polarity to be applied to the first electrode **704** and the determined second voltage and/or polarity to be applied to the second electrode **706**.

(155) In one embodiment, the sense circuit **2104** may include a voltage sense circuit **2104V** that is configured to sense a voltage potential measurement representative of a voltage difference/potential between the first voltage applied to the first electrode **704** and the second voltage applied to the second electrode **706**. The voltage sense circuit **2104V** is configured to sense a characteristic (e.g.,

voltage difference/potential between the first voltage applied to the first electrode **704** and the second voltage applied to the second electrode **706**) of the electrostatic clutch assembly **718**. The voltage sense circuit **2104V** is configured to provide the voltage potential measurement as an input to the control circuit **750**. That is, the sensing signal is transmitted from the electrostatic clutch assembly **718** to the control circuit **750** via the voltage sense circuit **2104V**. The control circuit **750** is configured to determine, based on the input from the voltage sense circuit **2104V**, the first voltage and/or polarity to be applied to the first electrode **704** and the second voltage and/or polarity to be applied to the second electrode **706**.

(156) In one embodiment, the sense circuit **2104** may include a current sense circuit **2104C** that is configured to sense a current measurement representative of a current flow between the first electrode **704** and the second electrode **706**. The current sense circuit **2104C** is configured to sense a characteristic (e.g., current flow between the first electrode **704** and the second electrode **706**) of the electrostatic clutch assembly **718**. The current sense circuit **2104C** is configured to provide the current measurement as an input to the control circuit **750**. That is, the sensing signal is transmitted from the electrostatic clutch assembly **718** to the control circuit **750** via the current sense circuit **2104C**. The control circuit **750** is configured to determine, based on the input from the current sense circuit **2104C**, the first voltage and/or polarity to be applied to the first electrode **704** and the second voltage and/or polarity to be applied to the second electrode **706**.

(157) In another embodiment, the sense circuit **2104** may include a motor speed sense circuit **2104S** that is configured to sense a motor rotation speed measurement. The motor rotation speed measurement from which acceleration, and angle may be derived. The motor speed sense circuit **2104S** is configured to sense a characteristic (e.g., motor rotation speed) of the motor **715** of the motor and transmission assembly **714**. The motor speed sense circuit **2104S** is configured to provide the motor rotation speed measurement as an input to the control circuit **750**. That is, the sensing signal is transmitted from the motor and transmission assembly **714** to the control circuit **750** via the sense circuit **2104S**. The control circuit **750** is configured to determine, based on the input from the sense circuit **2104S**, the first voltage and/or polarity to be applied to the first electrode **304** and the second voltage and/or polarity to be applied to the second electrode **706**.

(158) In another embodiment, referring to FIGS. **21** and **23**, the sense circuit **2104** may include a torque setting sense circuit **2104TS** that is configured to sense a torque setting interface measurement. U.S. Pat. No. 10,220,500, which is incorporated by reference in the patent application in its entirety, provides an exemplary torque setting interface for an electronic clutch. A similar type of torque setting interface could be implemented into the electrostatic clutch assembly **718** of the present patent application.

(159) The torque setting sense circuit **2104TS** is configured to provide the torque setting interface measurement as an input to the control circuit **750**. The control circuit **750** is configured to determine, based on the input from the sense circuit **2104TS**, the first voltage and/or polarity to be applied to the first electrode **704** and the second voltage and/or polarity to be applied to the second electrode **706**.

(160) Referring to FIGS. **21** and **23**, a torque setting user interface is configured to convert numerical torque setpoints/threshold values to the voltage of the electrostatic clutch assembly **718**. For example, in one embodiment, the torque setting interface may include a clutch ring that is coupled to a potentiometer/rheostat. In another embodiment, the torque setting interface may include a thumbwheel that is coupled to a potentiometer/rheostat. In yet another embodiment, the torque setting interface may include at least one pushbutton and a display.

(161) In one embodiment, the control signal from the display of the torque setting interface to the DC high voltage power supply **2106** is an analog voltage. In another embodiment, the control signal from the display of the torque setting interface to the DC high voltage power supply **2106** is a pulse width modulated (PWM) signal. In yet another embodiment, the control signal from the display of the torque setting interface to the DC high voltage power supply **2106** is a digital code.

(162) In one embodiment, as shown in FIG. 23, if the torque setpoint/threshold value is “r,” then the voltage output (e.g., V.sub.out) by the DC high voltage power supply **2106** to the H-Bridge/switch circuit **2102** is calculated using equation (1) below:

$V_{out} = k \cdot r \cdot \text{sup.2} + e$ equation (1) where V.sub.out is the voltage output by the DC high voltage power supply **2106** to the H-Bridge/switch circuit **2102**; k is a constant; r is a numerical torque setpoint/threshold value set by the user at a torque setting interface; and e is an error.

(163) In yet another embodiment, referring to FIGS. 21 and 22, the sense circuit **2104** may include a reaction torque sense circuit **2104T** that is configured to sense a reaction torque measurement RTM. The reaction torque sense circuit **2104T** is configured to provide the reaction torque measurement RTM as an input to the control circuit **750**. The control circuit **750** is configured to determine an error using the reaction torque measurement RTM and an applied torque AT. The control circuit **750** is configured to determine, based on the error, the first voltage and/or polarity to be applied to the first electrode **704** and the second voltage and/or polarity to be applied to the second electrode **706**. That is, the first voltage and/or polarity to be applied to the first electrode **704** and the second voltage and/or polarity to be applied to the second electrode **706** is adjusted to correct for the error due to the reaction torque on the power drill **710**.

(164) In one embodiment, an accelerometer or rate gyroscope is configured to measure reaction torque at the operator/user. This reaction torque measurement is then used to correct errors in an application/applied torque. For example, an error is calculated by subtracting the measured reaction torque from the applied torque. That is, $\text{error} = (\text{Applied torque} - \text{Reaction torque})$. The error is a computed signed voltage that is then added to the setpoint of the electrostatic clutch assembly **718**.

(165) Electrical control signals ECS between the DC high voltage power supply **2106** and the control circuit **750** and the electrical control signals ECS from the control circuit **750** and the switching circuit/H-Bridge **2102** are shown in FIG. 21.

(166) Mechanical connection(s) MC between the motor and transmission assembly **714** (including the motor **715** and the transmission **716**) and the electrostatic clutch assembly **718** and also the mechanical connection(s) MC between the electrostatic clutch assembly **718** and the end effector **722** are also shown in FIG. 21.

(167) Sensing signals SS between the electrostatic clutch assembly **718** and the control circuit **750** and also the sensing signals SS between the motor and transmission assembly **714** (including the motor **715** and the transmission **716**) and the control circuit **750** are shown in FIG. 21.

(168) Electrical power signals EPS from the power source **702** to the DC high voltage power supply **2106**, from the DC high voltage power supply **2106** to the switch circuit/H-Bridge **2102**, from the switch circuit/H-Bridge **2102** to the first and second electrodes **704**, **706** of the electrostatic clutch assembly **18**, from the power source **702** to the control circuit **750**, and from the control circuit **750** to the motor and transmission assembly **714** (including the motor **715** and the transmission **716**) are also shown in FIG. 21.

(169) In one embodiment, referring to FIG. 24, a method of operating the power tool **10** is provided. As discussed in various embodiments of the present patent application, the power tool **10** includes the motor **15**, the electrostatic clutch assembly **18**, the output spindle **68**, and the control circuit **50**. The electrostatic clutch assembly **18** includes the first electrode **304**, the second electrode **306** and the dielectric layer **308** arranged between the first electrode **304** and the second electrode **306**. The electrostatic clutch assembly **18** is positioned between the motor **15** and the output spindle **68**.

(170) The method of operating the power tool **10** includes actuating the motor **15**, and applying a first voltage difference between the first electrode **304** and the second electrode **306**. For example, in one embodiment, applying a first voltage to the first electrode **304** and a second voltage to the second electrode **306**, where the first voltage is different from the second voltage. The method of operating the power tool **10** also includes controlling the motor **15** to increase a torque applied to the output spindle **68** until the torque applied to the output spindle **68** reaches a first threshold

value, where one of the first electrode **304** and the second electrode **306** rotationally slips relative to the other of the first electrode **304** and the second electrode **306** when the torque on the output spindle **68** exceeds the first threshold value. The method of operating the power tool **10** further includes discharging the first voltage from the first electrode **304** and the second voltage from the second electrode **306**. The first threshold value may be interchangeably referred to as setpoint in FIG. **24**. The torque applied to the output spindle **68** may be substantially same as the torque on one of the first electrode and the second electrode **304** and **306**.

(171) The method of operating the power tool **10** also includes initiating a protective action upon detecting that one of the first electrode **304** and the second electrode **306** rotationally slips relative to the other of the first electrode **304** and the second electrode **306**. The initiating a protective action may include discharging the first voltage from the first electrode **304** and the second voltage from the second electrode **306**. The initiating a protective action may include reducing or interrupting power delivery to the motor **15**. The initiating a protective action may include braking the motor **15**. The first threshold value may correspond to a holding force between the first electrode **304** and a second electrode **306**. The holding force may correspond to a magnitude of a difference between the first voltage and the second voltage. The holding force is proportional to a square of the magnitude of the difference between the first voltage and the second voltage.

(172) FIG. **24** shows a graphical illustration of an embodiment of behavior of the power tool system **10** in operation. Applied torque (i.e., measured in Nm) is shown on left-hand side Y-axis of the graph, while motor speed (i.e., measured in revolutions/minute (RPM)) is shown on the right-hand side Y-axis of the graph. Time (i.e., measured in seconds) is shown in the X-axis of the graph.

(173) Referring to FIG. **24**, at time A, the motor **15** is actuated. The motor **15** may be actuated by pulling/actuating the trigger assembly **24**. When actuated/pulled the trigger assembly **24**, for example, cooperates with the battery pack **26** to provide power to the motor and transmission assembly **14** (including the motor **15**) in a manner that is generally well-known to a person of ordinary skill in the art. Also, from time A to time B, the control circuit **50** is configured to cause the first voltage is applied to the first electrode **304** and the second voltage is applied to the second electrode **306**, where the first voltage is different from the second voltage, generating a first attractive force between the first and second electrodes **304**, **306**.

(174) From time B to time E, the motor **15** is controlled (e.g., by actuating the trigger assembly **24**) to increase a torque applied to the output spindle **68** until the torque applied to the output spindle **68** reaches a first threshold value, where one of the first electrode **304** and the second electrode **306** rotationally slips relative to the other of the first electrode **304** and the second electrode **306** when the torque on the output spindle **68** exceeds the first threshold value. Specifically, from time B to time C, the motor **15** ramps up to a full speed with very little torque is applied to the fastener. From time C to time E, fastener torque is increased and the motor speed of the motor **15** decreases. From time C to time E, the change in the motor speed of the motor **15** is equivalent to the changed in the motor current of the motor **15**.

(175) At time E, the motor speed of the motor **15** indicates that torque on the output spindle **68** has almost reached desired setpoint (i.e., the first threshold value). At this time E, the motor control circuit causes the motor speed to decrease so that the torque increases more gradually. At time F, the electrostatic clutch assembly **18** slips (i.e., the torque on the output spindle **68** exceeded the first threshold value), causing the motor speed to briefly increase. This brief increase in current can be detected by the current sense circuit, which can communicate to the controller that the clutch has reached its slip torque, At time F, the applied torque does not immediately drop to zero because there is still some non-static friction between the first and second electrodes **304**, **306** of the electrostatic clutch assembly **18**.

(176) At time G, once slip of the clutch is detected, the controller causes the first and second electrodes **304**, **306** of the electrostatic clutch assembly **18** to be discharged (e.g., by shorting them across the H-bridge) and the motor speed of the motor **15** and the applied torque both quickly drop

to zero.

(177) After time G, the motor **15** may be controlled for providing feedback to a user. For example, after time G, in one embodiment, the motor **15** may be pulsed to provide haptic feedback to the user. In another embodiment, the haptic feedback to the user may be provided by a vibrator. Alternatively, or in addition, a visual feedback or an acoustic/a sound feedback may be provided to the user.

(178) FIGS. **28A** and **28B** show various exemplary indicators of slip indicating achieved torque setting. For example, as shown in FIG. **28A**, the indicators may be configured to provide visual feedback to the user, including, switching the light(s) **2801** on and off, change in color of the light(s) **2801**, change in the brightness of the light(s) **2801**, flashing/strobing of the light(s) **2801**, etc. As shown in FIG. **28B**, the indicators may include other visual indicators such as progress meter **2803** indicating the progress of the operation of the power tool **10**. Such progress meter **2803** may include different colored light output (e.g., green, yellow and red) that are turned on based on and to indicate the progress of the operation of the power tool **10**. Green colored light output **2805** may be turned on until a torque applied to the output spindle **68** below a first threshold value, yellow colored light output **2807** may be turned on when the torque applied to the output spindle **68** is approaching the first threshold value, and red colored light output **2809** may be turned on when the torque applied to the output spindle **68** exceeds the first threshold value.

(179) After, providing the feedback (e.g., pulsed/haptic, visual, acoustic/sound) to the user, the trigger assembly **24** of the power tool **10** may be released.

(180) The operation of the electrostatic clutch assembly **18** in a torque-limited mode is shown in FIG. **27**. FIG. **27** shows a graphical illustration of the operation of the electrostatic clutch assembly **18** in the torque-limited mode. Voltage (i.e., measured in volts) is shown on the left-hand side Y-axis, while time (i.e., measured in seconds) is shown in the X-axis of the graph. A voltage potential is applied to the electrostatic clutch assembly **18** between the plurality of electrodes **304** (e.g., disc stack A) and the plurality of electrodes **306** (e.g., disc stack B). In one embodiment, the plurality of electrodes may be referred to as disc stack (e.g., a stack of disc electrodes), the plurality of second electrodes **306** may be referred to as disc stack B, and the plurality of second electrodes **304** may be referred to as disc stack A in the discussions for FIGS. **27** and **29-31**.

(181) The voltage potential may include a voltage difference between voltage V1 that is applied to the plurality of electrodes **306** (e.g., disc stack B) and voltage V2 that is applied to the plurality of electrodes **304** (e.g., disc stack A). As shown in FIG. **27**, the voltage V1 that is applied to the plurality of electrodes **306** (e.g., disc stack B) is different from the voltage V2 that is applied to the plurality of electrodes **304** (e.g., disc stack A).

(182) The higher the voltage potential between the plurality of electrodes **304** (e.g., disc stack A) and the plurality of electrodes **306** (e.g., disc stack B), the greater the frictional torque coupling between the plurality of electrodes **304** (e.g., disc stack A) and the plurality of electrodes **306** (e.g., disc stack B). The voltage potential applied to the plurality of electrodes **304** (e.g., disc stack A) and the plurality of electrodes **306** (e.g., disc stack B) is proportional to the frictional coupling between the plurality of electrodes **304** (e.g., disc stack A) and the plurality of electrodes **306** (e.g., disc stack B), and thus a breakaway torque may be configured for the system.

(183) In the power tool **10**, the controller/control circuit **50** may be configured to adjust the DC amplitude of the high voltage power supply **2106**. This allows the user to select a breakaway torque setpoint. As torque builds on the output shaft/spindle **68** in the power tool **10**, the frictional coupling is eventually overcome causing a slip between the plurality of electrodes **304** (e.g., disc stack A) and the plurality of electrodes **306** (e.g., disc stack B). As a slip is detected, the voltage potential between the plurality of electrodes **304** (e.g., disc stack A) and the plurality of electrodes **306** (e.g., disc stack B) is immediately removed, decoupling the output shaft/spindle **68** from the motor **15** allowing the output shaft/spindle **68** to coast. This allows the electrostatic clutch assembly **18** to be operated as a torque limiting power tool (i.e., a torque wrench), or as a virtual

shear-pin in a machine as a safety device.

(184) The operation of the electrostatic clutch assembly **18** in an AC anti-polarization mode is shown in FIG. **29**. FIG. **29** shows a graphical illustration of reversing polarity across the electrodes **304**, **306** of the electrostatic clutch assembly **18** in alternating cycles of operation. Voltage (i.e., measured in volts) is shown on the left-hand side Y-axis, while time (i.e., measured in seconds) is shown in the X-axis of the graph.

(185) Certain materials used in the electrostatic clutch assembly **18** provide favorable attributes, but accumulate polarizing features limiting their function, after repeated operation in the same voltage polarity. Periodic reversal of the operational voltage polarity mitigates accumulated surface polarization. Reversal of operational polarity can be achieved using the switch circuit/H-Bridge **2102** to periodically transpose the voltage on the first and second electrodes **304** and **306** of the electrostatic clutch assembly **18**. This is described in detail above in FIG. **21** and with respect to the operation of the switch circuit/H-Bridge **2102**.

(186) In one embodiment, in the power tool **10**, the operational voltage polarity reversal can be applied instantaneously and periodically during the operation of the power tool **10** while maintaining the potential (and thus the attractive force) between the the first and second electrodes **304** and **306** of the electrostatic clutch assembly **18**. In another embodiment, the operational voltage polarity reversal can be applied during the off time between pulls of the trigger assembly **24**, beginning each tool use cycle with an applied polarity opposite the previous.

(187) For example, as shown in FIG. **29**, from time A to time B, voltage that is applied to the second plurality of electrodes **306** (e.g., disc stack B) is zero, while voltage that is applied to the first plurality of electrodes **304** (e.g., disc stack A) is V2. This is the same from time C to time D.

(188) From time B to time C, voltage that is applied to the second plurality of electrodes **306** (e.g., disc stack B) is V1, while voltage that is applied to the first plurality of electrodes **304** (e.g., disc stack A) is zero. This is the same from time D to time E.

(189) In another embodiment, not shown, from time A to time B, voltage that is applied to the second plurality of electrodes **306** (e.g., disc stack B) is V1, while voltage that is applied to the first plurality of electrodes **304** (e.g., disc stack A) is zero. This is the same from time C to time D.

(190) From time B to time C, voltage that is applied to the second plurality of electrodes **306** (e.g., disc stack B) is zero, while voltage that is applied to the first plurality of electrodes **304** (e.g., disc stack A) is V2. This is the same from time D to time E.

(191) The operation of the electrostatic clutch assembly **18** in an anti-cam-out mode is shown in FIG. **30**. FIG. **30** shows a graphical representation of the operation of the electrostatic clutch assembly **18** in the anti-cam-out mode. That is, FIG. **30** shows a graphical representation of short intervals of slip of the electrostatic clutch assembly **18** followed by re-engagement to prevent a fastener bit from camming out of or stripping the fastener head. Voltage (i.e., measured in volts) is shown on the left-hand side Y-axis of the graph, while time (i.e., measured in seconds) is shown in the X-axis of the graph.

(192) The electrostatic clutch assembly **18** may be operated with controlled modulation while driving a screw/fastener so as to prevent cam-out. The electrostatic clutch assembly **18** may be driven at a torque setpoint, while using the control circuit **50** to monitor the motor speed (e.g., see M3 using **2104/2104S** in FIG. **21**) of the motor **15** for a sudden acceleration. Sudden acceleration of the motor **15** (in the absence of a detected slip of the electrostatic clutch assembly **18**) while driving a fastener load is indicative of cam-out (i.e., slipping and displacement of the drive bit against the screw head). If the initiation of a cam-out event is detected, the clutch control voltage is lowered slightly using the setpoint of the programmable power supply, allowing the electrostatic clutch assembly **18** to slip (i.e., rather than at the bit-to-fastener interface). This intentional momentary slip of the electrostatic clutch assembly **18** is configured to allow the relief of the cam-out forces, correcting the engagement of the tool bit to the fastener. This process can be repeated multiple times as necessary in the driving of a fastener, allowing aggressive driving of a fastener

without damage.

(193) In FIG. 30, the voltage that is applied to the plurality of electrodes 304 (e.g., disc stack A) is controllably modulated while driving a screw/fastener so as to prevent cam-out. That is, the voltage that is applied to the first plurality of electrodes 304 (e.g., disc stack A) is high and at voltage V2 from time A to time B and from time C to time D and is low or zero at voltage V3 from time B to time C and from time D to time E such that the average voltage $V_{sub.A}$ is applied to the first plurality of electrodes 304 (e.g., disc stack A). Also, as shown in FIG. 30, voltage V1 is applied to the second plurality of electrodes 306 (e.g., disc stack B) and may be maintained constant.

(194) In another embodiment, not shown, as would be appreciated by a person of ordinary skill in the art, the voltage that is applied to the second plurality of electrodes 306 (e.g., disc stack B) may be controllably modulated while driving a screw/fastener so as to prevent cam-out. That is, the voltage that is applied to the second plurality of electrodes 306 (e.g., disc stack B) is high and at voltage V1 from time A to time B and from time C to time D and is low or zero at voltage V3 from time B to time C and from time D to time E such that the average voltage (e.g., (not shown)) is applied to the plurality of electrodes 306 (e.g., disc stack B). In such an embodiment, voltage V2 is applied to the first plurality of electrodes 304 (e.g., disc stack A) and may be maintained constant.

(195) In yet another embodiment, the voltage applied to the electrostatic clutch assembly 18 may also be controlled by varying the width of the applied pulses and thereby varying the average voltage $V_{sub.A}$ applied to the electrodes 304, 306 of the electrostatic clutch assembly 18.

(196) The operation of the electrostatic clutch assembly 18 in an impact mode is shown in FIG. 31. FIG. 31 shows a graphical representation of the operation of the electrostatic clutch assembly 18 in the impact mode. That is, FIG. 31 shows a graphical representation of short intervals of slip of the electrostatic clutch assembly 18 followed by re-engagement to simulate operation of an impact mechanism in a direct drive power tool. Voltage (i.e., measured in volts) is shown on the left-hand side Y-axis of the graph, while time (i.e., measured in seconds) is shown in the X-axis of the graph.

(197) The electrostatic clutch assembly 18 may be repeatedly engaged and disengaged while driving a fastener, creating a virtual impact mechanism under electronic control. This may be accomplished by using the H-Bridge/switch circuit 2102 to fully remove and restore the voltage potential on the first electrodes 304 (i.e., disc stack A) and the second electrodes 306 (i.e., disc stack B) of the electrostatic clutch assembly 18 repeatedly with a given period and duty cycle. This impact mode could likewise be employed to create a virtual hammer drill mechanism, by limiting the off time of the duty cycle to something significantly less than 50% (i.e., conserving forward momentum, while still introducing an impulse event).

(198) In FIG. 31, the voltage that is applied to the plurality of electrodes 304 (e.g., disc stack A) is fully removed and restored repeatedly with a given period and duty cycle. For example, a voltage V2 is applied to the plurality of first electrodes 304 (e.g., disc stack A) from time A to time B and from time C to time D, while the voltage is reduced or fully removed from the first plurality of electrodes 304 (e.g., disc stack A) from time B to time C and from time D to time E. As shown in FIG. 31, the voltage that is applied to the second plurality of electrodes 306 (e.g., disc stack B) is maintained constant, for example, at zero.

(199) In another embodiment, not shown, the voltage that is applied to the second plurality of electrodes 306 (e.g., disc stack B) may be fully removed and restored repeatedly with a given period and duty cycle. For example, a voltage V1 is applied to the second plurality of electrodes 306 (e.g., disc stack B) from time A to time B and from time C to time D, while the voltage is fully removed from the first plurality of electrodes 304 (e.g., disc stack B) from time B to time C and from time D to time E. In such an embodiment, the voltage that is applied to the plurality of electrodes 304 (e.g., disc stack A) is maintained constant.

(200) In one embodiment, an electric field above a first predetermined threshold is applied across the first electrode 304 and the second electrode 306 causing a first electrostatic force between the first electrode 304 and the second electrode 306 to rotationally couple the shaft 302 and the clutch

housing **310** of the electrostatic clutch assembly **18** such that the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** rotate together at the same velocity. An electric field below a second predetermined threshold between the first electrode **304** and the second electrode **306** causes one of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18** to be rotationally decoupled from the other of the shaft **302** and the clutch housing **310** of the electrostatic clutch assembly **18**. An electric field between the first predetermined threshold and the second predetermined threshold is applied across the first electrode **304** and the second electrode **306** causing a second electrostatic force between the first electrode **304** and the second electrode **306** to rotationally couple the output member **68** with the motor and transmission assembly **14** such that the output member **68** rotates together at the same velocity when the torque therebetween is below a set torque and to decouple the output member **68** from the motor and transmission assembly **14** so that the output member **68** can rotate at a lower or zero velocity while the motor and transmission assembly **14** continues to rotate at a higher velocity when the torque therebetween is above the set torque.

(201) When the electrostatic clutch assembly **18** is in a clutch mode and when the torque between the output shaft **68** and the motor and transmission assembly **14** is above the set torque, the velocity of the output shaft **68** is zero. When the electrostatic clutch assembly **18** is in the clutch mode, the first electrode **304** and the second electrode **306** of the electrostatic clutch assembly **18** are variably attracted to each other. When the electrostatic clutch assembly **18** is in the clutch mode, the first electrode **304** and the second electrode **306** of the electrostatic clutch assembly **18** are positioned in such a way that no gap (i.e., only dielectric material layer **308**) exists between surfaces of the first electrode **304** and the second electrode **306** that face each other.

(202) In one embodiment, the clutch mode of the electrostatic clutch assembly **18** may include a first mode of operation and a second mode of operation.

(203) In the first mode of operation, the control circuit **50** causes a first voltage to be applied across the first electrode **304** and a second voltage with a polarity opposite a polarity the first voltage to be applied to the second electrode **306**, generating a first attractive force between the first and second electrodes **304**, **306**, which causes the output member **68** to rotate together with the motor and transmission assembly **14** when a torque on the output member **68** is less than or equal to a first threshold value (e.g., when the torque on the output member is such that a shear force or torque about the rotational axis of the output member is less than or equal to the attractive force between the electrodes **304**, **306**) and which causes one of the shaft **302** and the clutch housing **310** to rotationally slip relative to the other of the shaft **302** and the clutch housing **310** when the torque on the output member **68** exceeds the first threshold value (e.g., when the torque on the output member is such that a shear force or torque about the rotational axis of the output member is greater than the attractive force between the electrodes), interrupting torque transmission from the motor and transmission assembly **14** to the output member **68**. In one embodiment, the first threshold corresponds to the first attractive force.

(204) In one embodiment, in the second mode of operation, the control circuit **50** causes a third voltage to be applied across the first electrode **304** and a fourth voltage with a polarity opposite a polarity the third voltage to be applied to the second electrode **306**, generating a second attractive force between the first and second electrodes **304**, **306**, which causes one of the shaft **302** and the clutch housing **310** to rotate together with the other of the shaft **302** and the clutch housing **310** when a torque on the output member **68** is less than or equal to a second threshold value and which causes one of the shaft **302** and the clutch housing **310** to (e.g., rotationally) slip or rotate relative to the other of the shaft **302** and the clutch housing **310** when the torque on the output member **68** exceeds the second threshold value, interrupting torque transmission from the input member **66** to the output member **68**. That is, in one embodiment, in the second mode of operation, the control circuit **50** causes the third voltage to be applied to the first electrode **304** and a different fourth voltage to be applied to the second electrode **306**.

(205) In one embodiment, the third voltage is greater than the first voltage, the fourth voltage is greater than the second voltage, the second attractive force is greater than the first attractive force, and the second threshold value is greater than the first threshold value. In one embodiment, a second voltage difference between the third voltage and the fourth voltage is greater than a first voltage difference between the first voltage and the second voltage. The power tool **10** may further include a selector switch coupled to the housing **12** that is actuatable by a user to select between the first and second modes of operation.

(206) In one embodiment, in a third mode of operation, the control circuit **50** causes a zero voltage difference to be applied to the first and second electrodes **304, 306**, allowing the second electrode to rotate relative to the first electrode and preventing torque transmission from one of the shaft **302** and the clutch housing **310** to the other of the shaft **302** and the clutch housing **310**. In one embodiment, the control circuit **50** is configured to automatically switch from the first mode to the third mode upon sensing (e.g., by a sensor) that one of the shaft **302** and the clutch housing **310** has rotationally slipped or rotated relative to the other of the shaft **302** and the clutch housing **310**.

(207) In one embodiment, in a fourth mode of operation, the electrostatic clutch assembly **18** is configured to prevent interruption of torque transmission from one of the shaft **302** and the clutch housing **310** to the other of the shaft **302** and the clutch housing **310**.

(208) In one embodiment, in the fourth mode of operation, the control circuit may cause a fifth voltage to be applied across the first electrode **304** and a different sixth voltage to be applied to the second electrode **306**, generating a third attractive force between the first and second electrodes **304, 306**, the third attractive force exceeding a torque on the output member **68** during operation of the power tool **10**. The second voltage may have a polarity opposite a polarity of the first voltage. In one embodiment, the first voltage difference is user selectable to adjust the first attractive force and the first threshold value. In one embodiment, a greater voltage difference corresponds to a greater first attractive force and a greater first threshold value. In one embodiment, the electrostatic clutch assembly **18** is held in the clutch mode such that when the working element meets a dynamic impact event, the electrostatic clutch assembly **18** slips or rotates. Thus, this configuration provides protection to the elements of the power tool, e.g., a mower blade striking a rock.

(209) When energized, the electrodes **304, 306** rotate together as a unit until the output torque on the chuck/tool holder **22** exceeds the holding force of the electrostatic films/layers. At this time, the electrodes **304, 306** rotate relative to one another with only a small frictional force between them. The input electrode **304, 306** will continue to rotate, while the output electrode **304, 306** coasts to rest.

(210) In one embodiment, the power tool **10** may further comprise a selector switch actuatable by a user to select between the first and second modes of operation. In a third mode of operation, the control circuit **50** causes zero voltage to be applied to the first and second electrodes **304, 306**, preventing motion from being transmitted from the motor and transmission assembly **14** to the output member **68**. In one embodiment, the control circuit **50** is configured to automatically switch from the first mode to the third mode upon sensing that the motion transmission from the motor and transmission assembly **14** to the output member **68** has been interrupted.

(211) Although the present patent application has been described in detail for the purpose of illustration, it is to be understood that such detail is solely for that purpose and that the present patent application is not limited to the disclosed embodiments, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. In addition, it is to be understood that the present patent application contemplates that, to the extent possible, one or more features of any embodiment can be combined with one or more features of any other embodiment.

Claims

1. A power tool comprising: a power tool housing; a motor received in the power tool housing and configured to be driven by a power source; an end effector configured to perform an operation on a workpiece; an electrostatic clutch assembly disposed in the power tool housing and between the motor and the end effector, the electrostatic clutch assembly comprising: a shaft; a first electrode and a second electrode axially arranged along an axial direction of the shaft, the first electrode and the second electrode being substantially parallel and adjacent to each other; a dielectric layer arranged between the first electrode and the second electrode; a clutch housing containing the first electrode, the second electrode, and at least a portion of the shaft; wherein the first electrode is non-rotatably coupled to the clutch housing and is axially movable relative to the clutch housing; wherein the second electrode is non-rotatably coupled to the shaft and is axially movable relative to the shaft; and a control circuit disposed in the power tool housing and operatively cooperable with the electrostatic clutch assembly, wherein the control circuit causes a first voltage difference to be applied between the first electrode and the second electrode, generating a first attractive force between the first and second electrodes, causing the shaft and the clutch housing to be coupled to each other when a torque on one of the shaft or the clutch housing is less than or equal to a first threshold value, and allowing the one of the shaft or the clutch housing to rotationally slip relative to the other of the shaft or the clutch housing when the torque on the one of the shaft or the clutch housing exceeds the first threshold value, wherein the end effector is mounted to an output spindle of the power tool, and wherein the power tool includes: the motor, and an impact mechanism that is configured to apply intermittent rotational impacts to the output spindle.
2. The power tool of claim 1, wherein the shaft and the clutch housing are non-rotatably coupled to each other when the torque on the shaft is less than or equal to the first threshold value.
3. The power tool of claim 1, wherein the first electrode is one of a plurality of first electrodes in the electrostatic clutch assembly, wherein the second electrode is one of a plurality of second electrodes in the electrostatic clutch assembly, wherein the first electrode and the second electrode form an electrode pair, wherein the electrode pair is one of a plurality of electrode pairs that are axially arranged along the axial direction of the shaft, and wherein the plurality of electrode pairs is substantially parallel to each other.
4. The power tool of claim 1, wherein each first electrode of the plurality of electrode pairs includes an annular plate member, and each second electrode of the plurality of electrode pairs includes an annular plate member.
5. The power tool of claim 3, wherein each first electrode of the plurality of electrode pairs has a first surface and a second surface and each second electrode of the plurality of electrode pairs has a first surface and a second surface that is facing the first surface of each first electrode.
6. The power tool of claim 5, wherein the dielectric layer is disposed on surfaces of the first electrode and the second electrode that face each other.
7. The power tool of claim 5, wherein the electrostatic clutch assembly further comprises a conductive material on at least one of the first surface of each first electrode or the second surface of each second electrode.
8. The power tool of claim 7, wherein the conductive material is disposed on surfaces of each first electrode and each second electrode that face each other.
9. The power tool of claim 3, wherein the end effector includes a tool holder that is configured to receive a tool bit portion therein, wherein the shaft has a first portion and a second portion, wherein one of the first portion and the second portion of the shaft is operatively connected to each second electrode of the plurality of electrode pairs, and wherein the other of the first portion and the second portion of the shaft is operatively connected to the output spindle of the power tool.
10. The power tool of claim 1, wherein the impact mechanism is a rotational impact mechanism.
11. The power tool of claim 10, wherein the impact mechanism comprises a Pott-type impact mechanism, and wherein the other of the shaft or the clutch housing is operatively coupled to an

output of the Pott-type impact mechanism.

12. The power tool of claim 11, wherein the Pott-type impact mechanism comprises a cam shaft rotationally driven by the motor, a hammer received at least partially over the cam shaft and operatively coupled to the cam shaft to be movable axially and rotationally relative to the cam shaft, an anvil, and a spring axially biasing the hammer toward the anvil, wherein, when the torque is at or below a threshold value, the cam shaft, the hammer, and the anvil rotate in unison, and wherein, when the torque exceeds the threshold value, the hammer moves axially and rotationally along the cam shaft to apply rotational impacts to the anvil, and wherein the other of the shaft or the clutch housing is operatively coupled to the anvil.

13. The power tool of claim 10, wherein the impact mechanism includes a twin hammer impact mechanism, and wherein the other of the shaft or the clutch housing is operatively coupled to an output of the twin hammer impact mechanism.

14. The power tool of claim 12, wherein the motor is an electric motor.

15. The power tool of claim 10, wherein the impact mechanism includes an oil pulse impact mechanism, and wherein the other of the shaft or the clutch housing is operatively coupled to an output of the oil pulse impact mechanism.

16. The power tool of claim 1, wherein the electrostatic clutch assembly is configured to receive an input torque from the impact mechanism and produce an output torque in response thereto, the output torque being limited to a maximum threshold value and transmitted to the output spindle of the power tool.

17. The power tool of claim 16, wherein the maximum threshold value is a fixed value.

18. The power tool of claim 16, wherein the maximum threshold value is configured to be adjusted dynamically.

19. The power tool of claim 16, wherein the maximum threshold value is a variable value.

20. The power tool of claim 16, wherein the maximum threshold value is a function of time.
